Electronics and Communication Systems
Electronics Systems

Master Degree in Computer Engineering

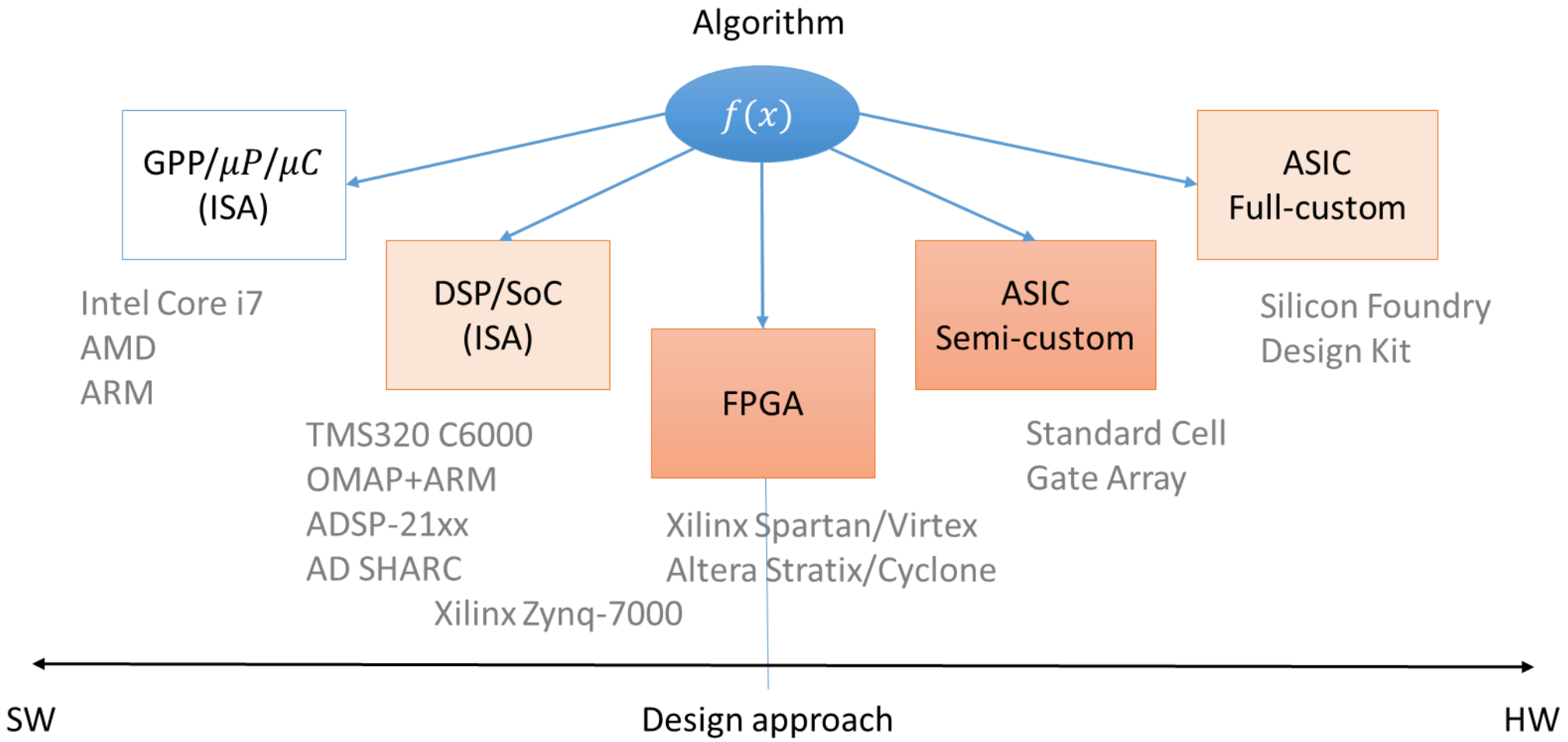
https://computer.ing.unipi.it/ce-lm

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Outline

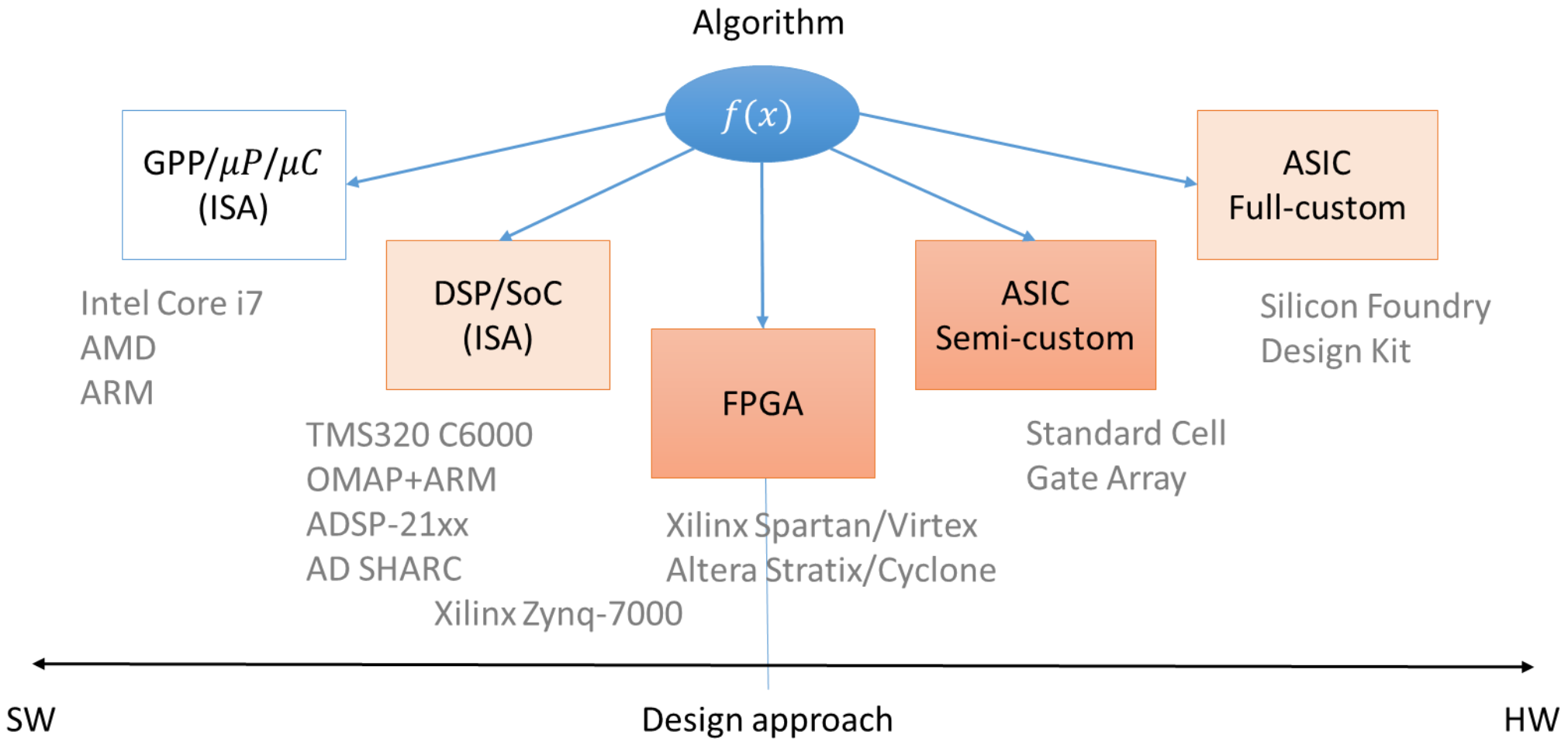
- Design Metrics
- Managing engineering trade-offs

How to map a system/algorithm to hw/sw design



Design is the art of managing engineering trade-offs !

How to map a system/algorithm to hw/sw design



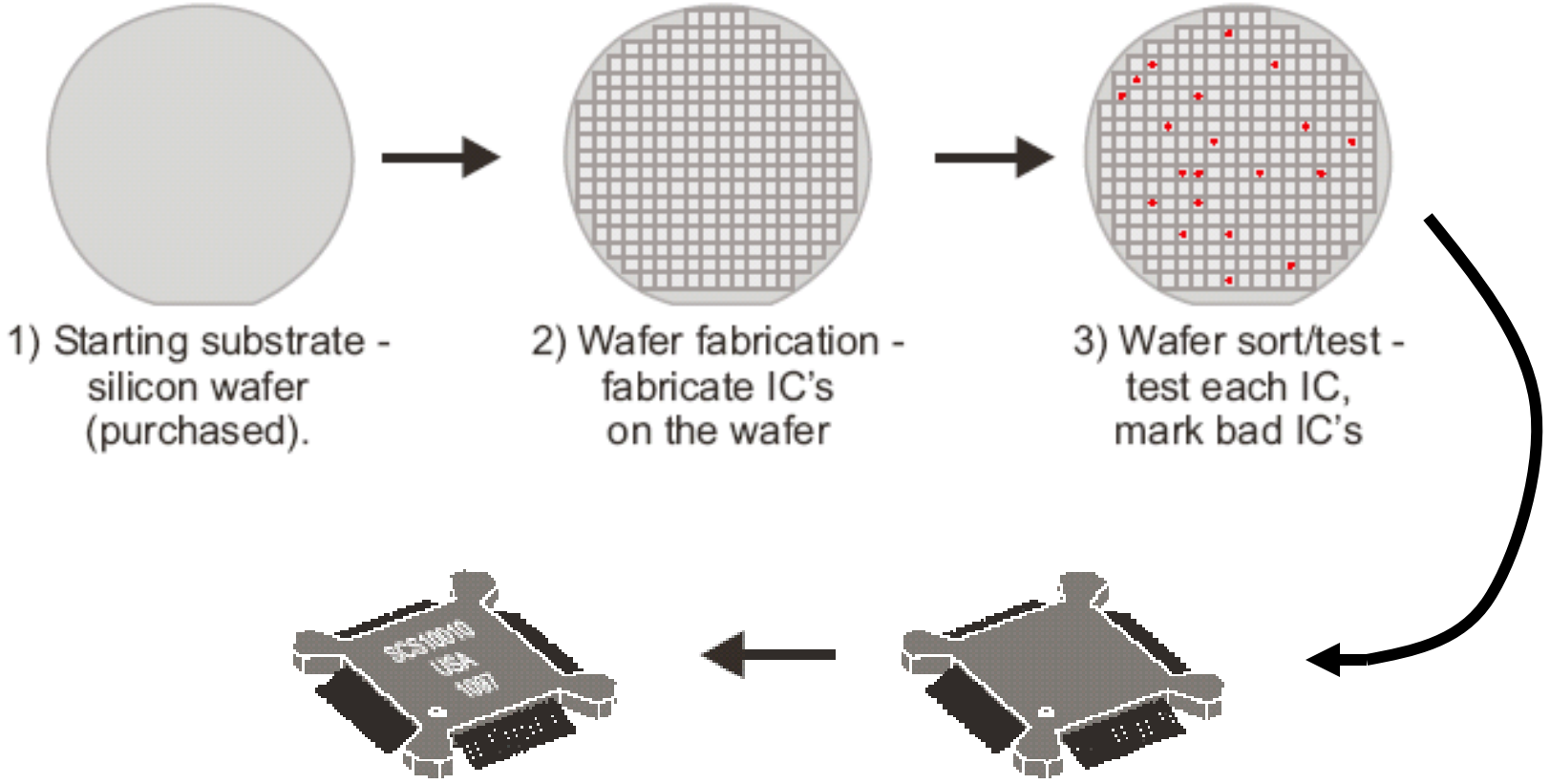
Let's see some fundamental design metrics !!

Fundamental Design Metrics

- Functionality
- Cost
  - NRE (fixed) costs - design effort
  - RE (variable) costs - cost of parts, assembly, test
- Reliability, robustness
  - Noise margins
  - Noise immunity
- Performance
  - Speed (delay)
  - Power consumption; energy
- Time-to-market

Design is the art of managing engineering trade-offs !

How a chip is manufactured?



1) Starting substrate - silicon wafer (purchased).

2) Wafer fabrication - fabricate IC's on the wafer

3) Wafer sort/test - test each IC, mark bad IC's

4) Packaging - assemble IC's into packages

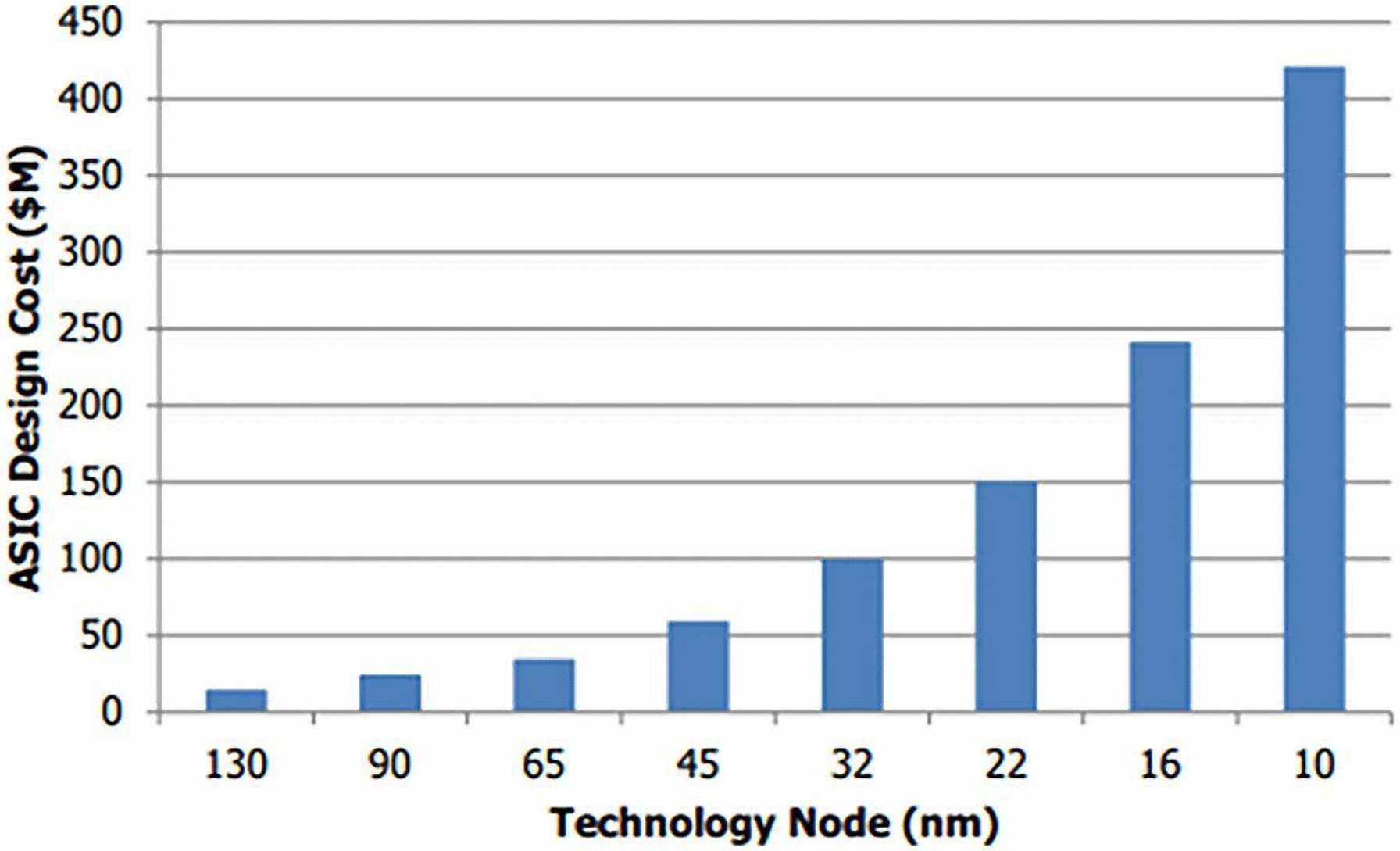
5) Mark & class/final test - mark and final test packaged product

Cost of Integrated Circuits

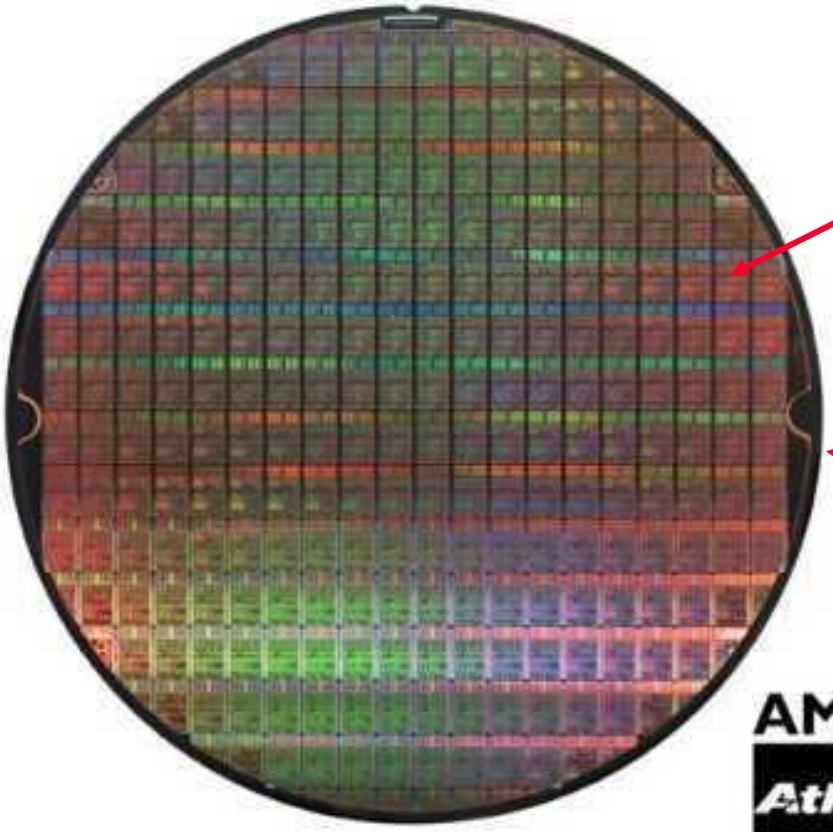
- NRE (non-recurring engineering) costs
  - Fixed cost to produce the design
    - design effort
    - design verification effort
    - mask generation
  - Influenced by the design complexity and designer productivity
  - More pronounced for small volume products
- Recurring costs – proportional to product volume
  - silicon processing
    - also proportional to chip area
  - assembly (packaging)
  - test

$$\text{cost per IC} = \text{variable cost per IC} + \frac{\text{fixed cost}}{\text{volume}}$$

NRE Cost is Increasing

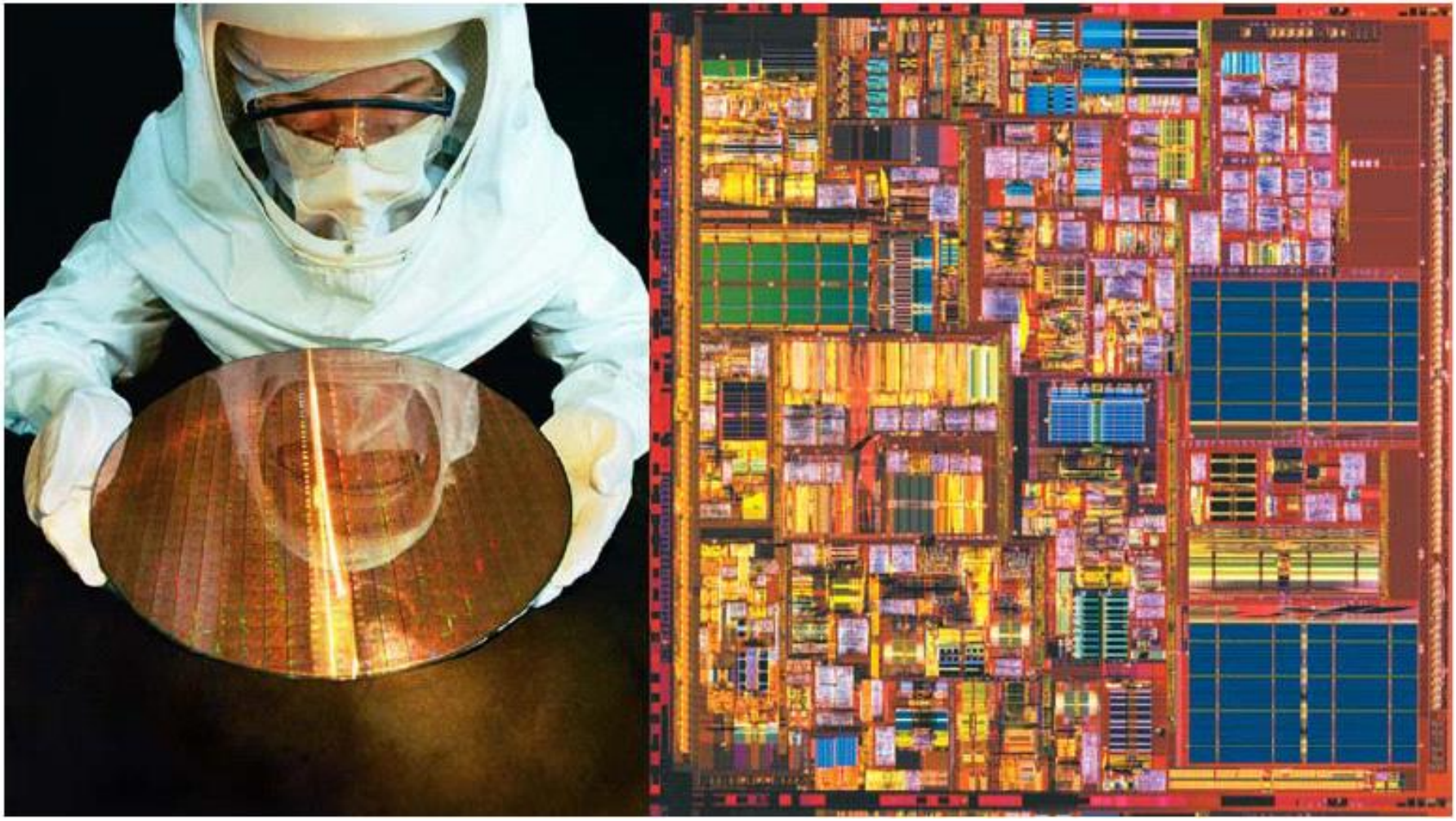


Silicon Wafer



From http://www.amd.com

Silicon Wafer



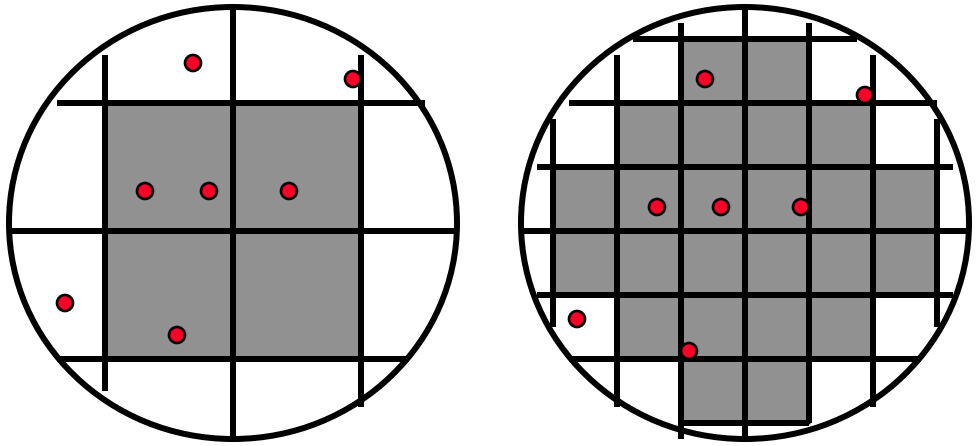
300mm wafer and Pentium 4 IC. Photos courtesy of Intel.

Recurring Costs

$$\text{variable cost} = \frac{\text{cost of die + cost of die test + cost of packaging}}{\text{final test yield}}$$

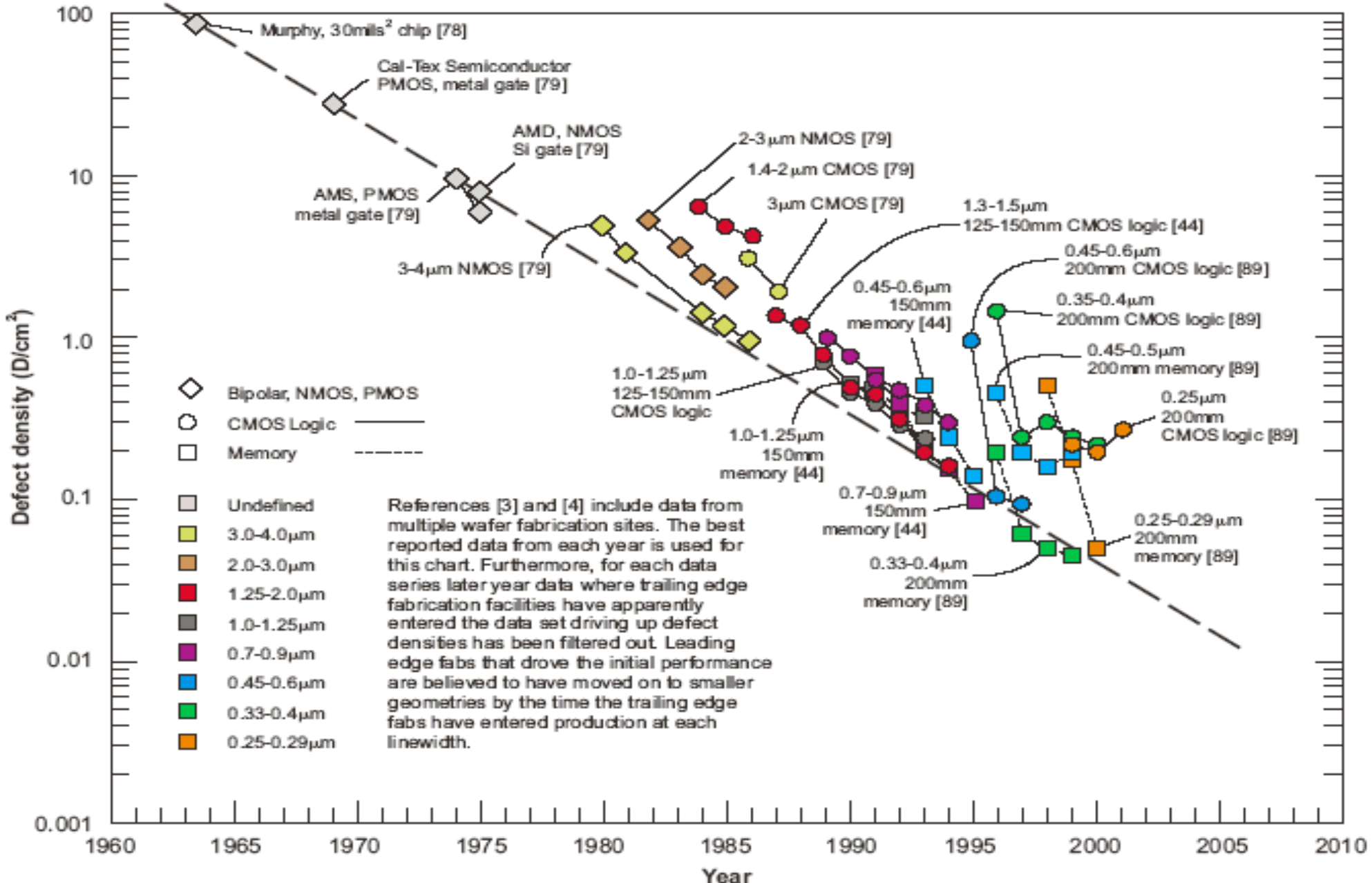
$$\text{cost of die} = \frac{\text{cost of wafer}}{\text{dies per wafer} \times \text{die yield}}$$

$$\text{dies per wafer} = \frac{\pi \times (\text{wafer diameter}/2)^2}{\text{die area}} - \frac{\pi \times \text{wafer diameter}}{\sqrt{2} \times \text{die area}}$$



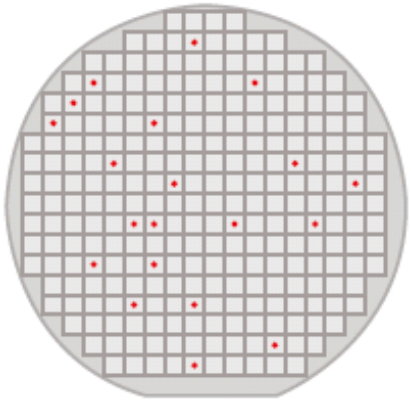
$$\text{die yield} = (1 + (\text{defects per unit area} \times \text{die area})/\alpha)^{-\alpha}$$

Defect Density Trends

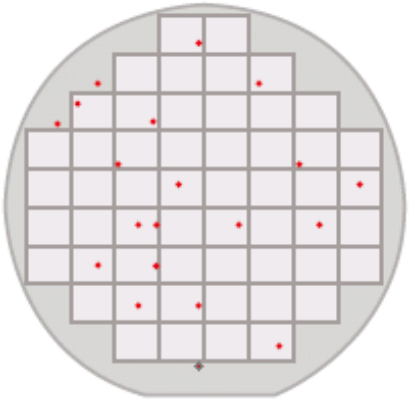


Yield Example

- Example
  - wafer size of 12 inches, die size of 2.5 cm^2, 1 defects/cm^2, α = 3 (measure of manufacturing process complexity)
  - 252 dies/wafer (remember, wafers round & dies square)
  - die yield of 16%
  - 252 x 16% = only 40 dies/wafer die yield !



20 Defects
20 Bad Die
264 Gross Die
92% Yield



20 Defects
16 Bad Die
54 Gross Die
70% Yield

- Die cost is strong function of die area
  - proportional to the third or fourth power of the die area

Examples of Cost Metrics (circa 1994)

Chip	Metal layers	Line width	Wafer cost	Defects /cm^2	Area (mm^2)	Dies/ wafer	Yield	Die cost
386DX	2	0.90	$900	1.0	43	360	71%	$4
486DX2	3	0.80	$1200	1.0	81	181	54%	$12
PowerPC 601	4	0.80	$1700	1.3	121	115	28%	$53
HP PA 7100	3	0.80	$1300	1.0	196	66	27%	$73
DEC Alpha	3	0.70	$1500	1.2	234	53	19%	$149
Super SPARC	3	0.70	$1700	1.6	256	48	13%	$272
Pentium	3	0.80	$1500	1.5	296	40	9%	$417



Examples of Wafer Cost (2003)

Wafer size	Line- width (μm)	Mask layers				
		18	20	22	24	26
200mm	0.25	$890	$980	$1,070	$1,155	-
	0.18	-	$1,320	$1,440	$1,565	-
	0.13	-	-	$1,815	$1,970	$2,130
300mm	0.13	-	-	$2,500	$2,690	$2,890
	0.09	-	-	-	$2,860	$3,065

Source: Icknowledge.com

Examples of Cost Metrics (2003) Intel Pentium4

Fab facility	200mm - 130nm - CMOS logic - 1 layer poly - 6 layer copper - FSG ILD					
Fab capacity (wafers/month)	30,000	United States				
Fab utilization	90%					
Wafer costs	Material	Labor	Overhead	Total		
Unyielded wafer costs	$85.00	$102.53	$1,511.78	$1,699.31		
Wafer yield	97.7%					
Yielded wafer costs	$87.00	$104.94	$1,547.33	$1,739.28		
	Area (mm^2)	Gross die	Die yield	Net die	Cost/die	
Die per wafer	146.0	177	70.2%	124	$14.03	
	Sort 1	Burn-in	Sort 2	Total		
Wafer sort costs	$0.939	$0.000	$0.000	$0.939		
Package	FCPGA - 35mm x 35mm x 478pins - Intel 130nm Pentium 4 with 512Kb cache					
	Package Cost	Yield	Yielded w die			
Packaging	$6.797	95.3%	$22.828			
	Class 1	Burn-in	Class 2	Total	Yield	Cost adder
Class test	$1.031	$0.080	$1.031	$2.142	97.7%	$0.000

Final cost: 25.57$ / chip source:www.icknowledge.com

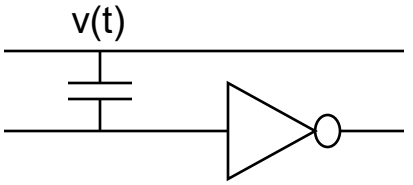
Fundamental Design Metrics

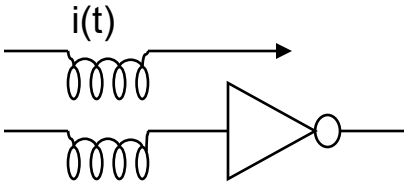
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- Cost
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- **Reliability, robustness**
  - Noise margins
  - Noise immunity
- Performance
  - Speed (delay)
  - Power consumption; energy
- Time-to-market

Reliability

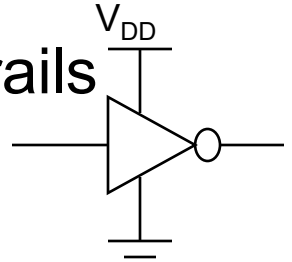
Noise in Digital Integrated Circuits

- Noise – unwanted variations of voltages and currents at the logic nodes
- From two wires placed side by side
  - capacitive coupling
    - voltage change on one wire can influence signal on the neighboring wire
    - cross talk
  - inductive coupling
    - current change on one wire can influence signal on the neighboring wire





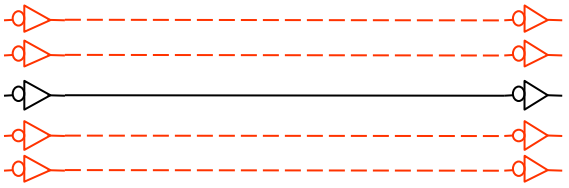
- From noise on the power and ground supply rails
  - can influence signal levels in the gate



Example of Capacitive Coupling

- Signal wire glitches as large as 80% of the supply voltage will be common due to crosstalk between neighboring wires as feature sizes continue to scale

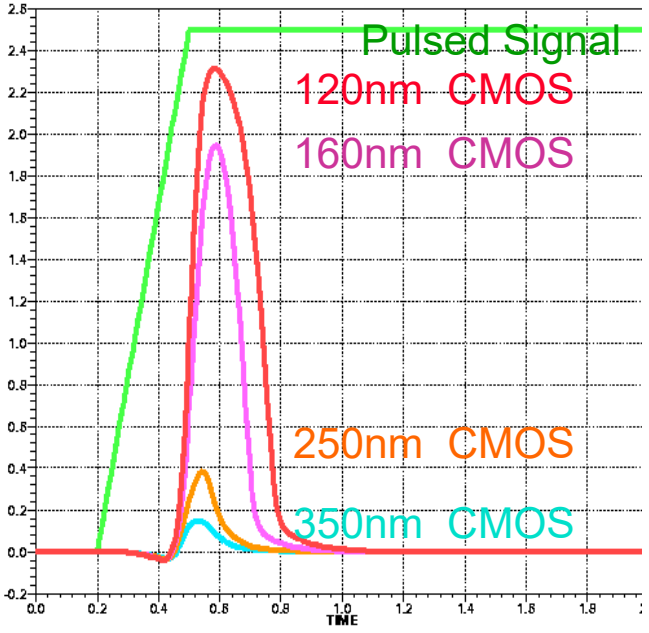
Crosstalk vs. Technology



Black line quiet

Red lines pulsed

Glitches strength vs technology



From Dunlop, Lucent, 2000

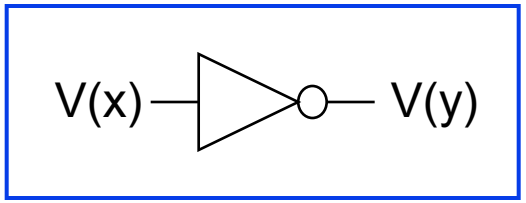
Static Gate Behavior

- Steady-state parameters of a gate – *static behavior* – tell how robust a circuit is with respect to both variations in the manufacturing process and to noise disturbances.
- Digital circuits perform operations on Boolean variables

$$x \in \{0,1\}$$

- A logical variable is associated with a *nominal voltage level* for each logic state

$$1 \Leftrightarrow V_{OH} \text{ and } 0 \Leftrightarrow V_{OL}$$



$V_{OH} = !\,(V_{OL})$

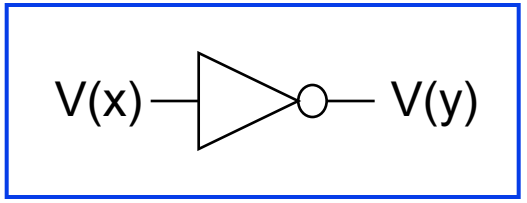
$V_{OL} = !\,(V_{OH})$

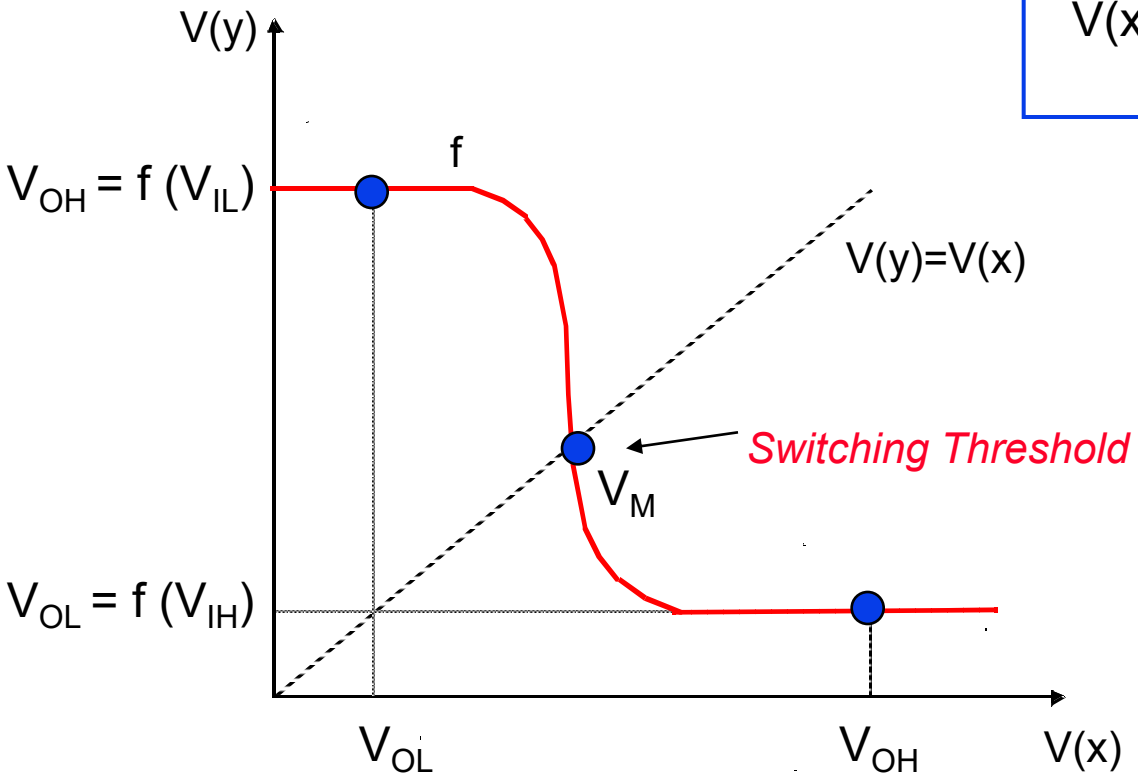
- Difference between V_{OH} and V_{OL} is the logic or *signal swing* $V_{sw} = V_{OH} - V_{OL}$

DC Operation

Voltage Transfer Characteristics (VTC)

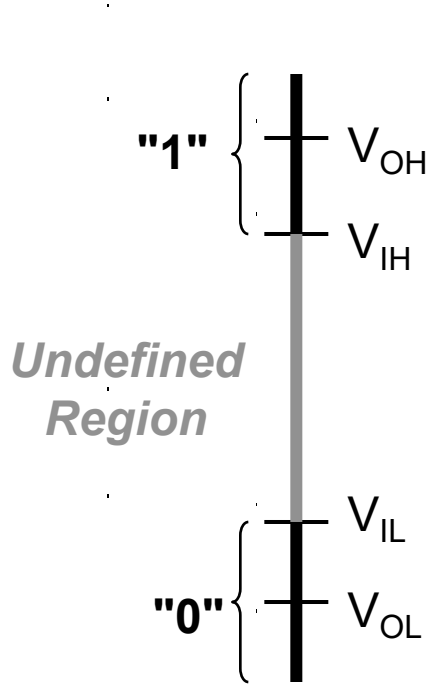
- Plot of output voltage as a function of the input voltage

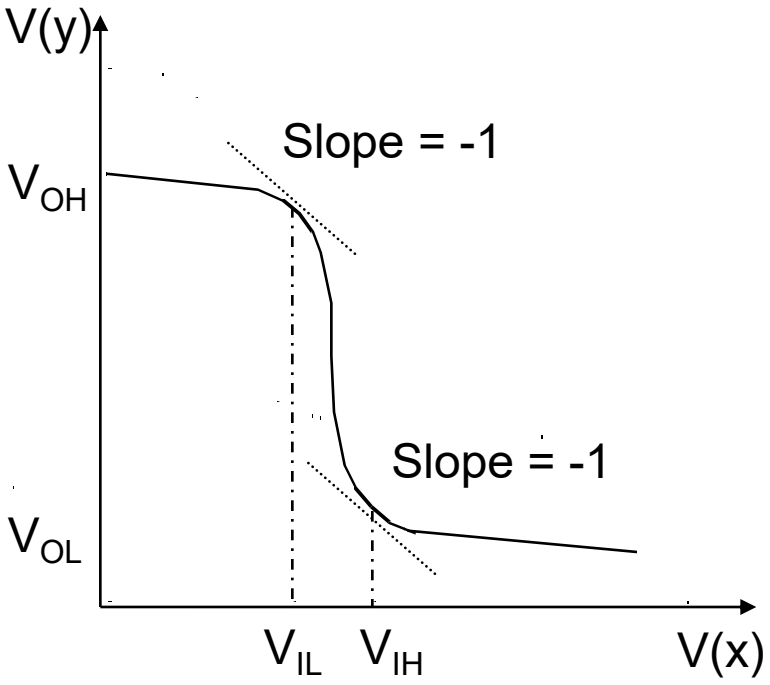




Mapping Logic Levels to the Voltage Domain

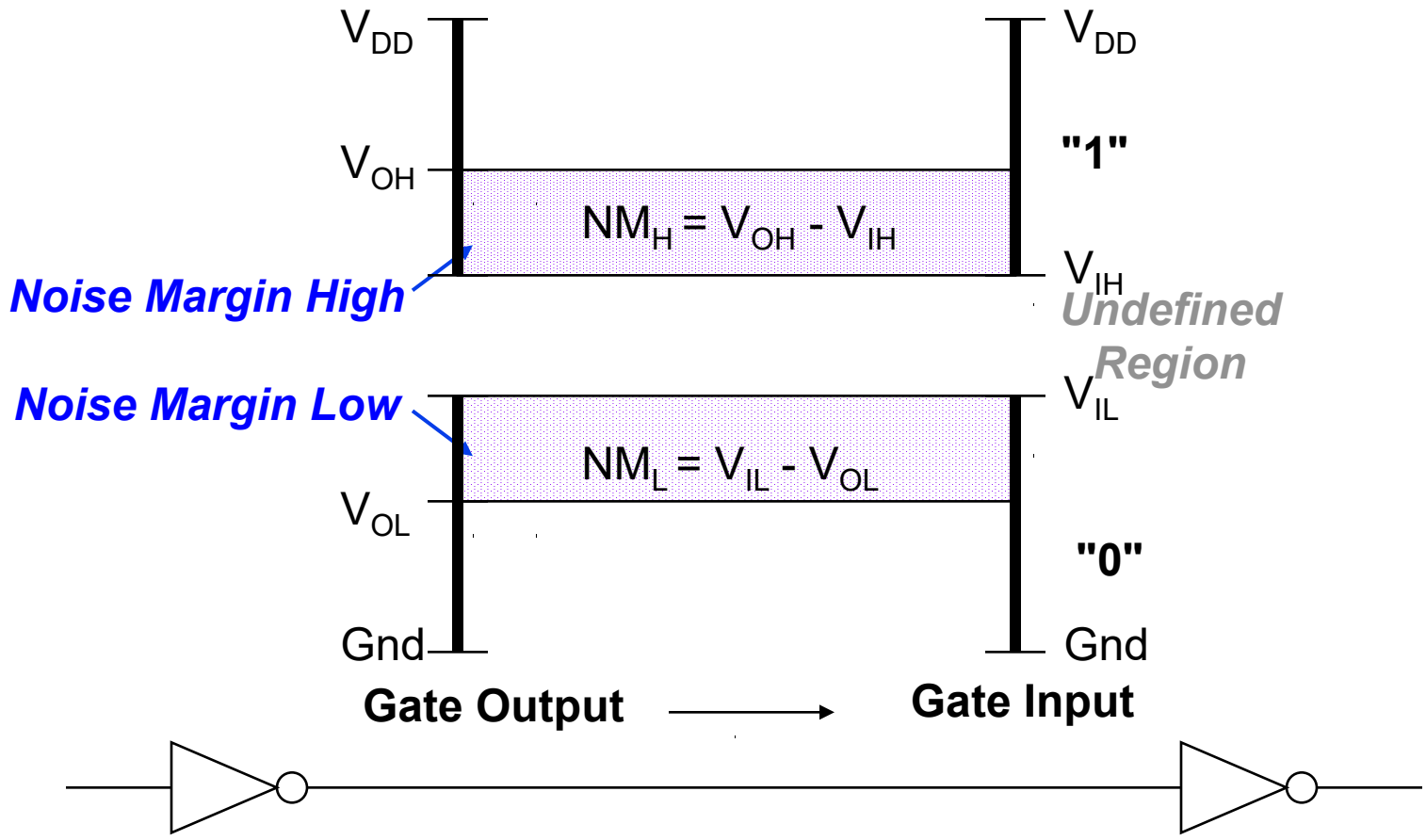
- The regions of acceptable high and low voltages are delimited by V_{IH} and V_{IL} that represent the points on the VTC curve where the gain = -1





Noise Margins

- For robust circuits, want the “0” and “1” intervals to be as large as possible



- Large noise margins are desirable, but not sufficient …

Noise Immunity

- Noise margin expresses the ability of a circuit to overpower a noise source
  - noise sources: supply noise, cross talk, interference, offset
- Absolute noise margin values are deceptive
  - a floating node is more easily disturbed than a node driven by a low impedance (in terms of voltage)
- *Noise immunity* expresses the ability of the system to process and transmit information correctly in the presence of noise
- For good noise immunity, the signal swing (i.e., the difference between V_{OH} and V_{OL}) and the noise margin have to be large enough to overpower the impact of fixed sources of noise

Case Example: The space radiation environment

- Principal effects due to galactic cosmic rays and solar activity:
  - Total Ionization Dose (TID): threshold voltage shift and leakage current in CMOS transistors (can lead to a complete functional failure)
  - Single Event Effects (SEE)
  - Single Event Latch-up (SEL): induce disruptive latch-up
  - Single Event Upset (SEU): temporary bit-flip that alters a datum but does not damage the electronic circuitry
- Need for rad-hard technology or radiation tolerant design

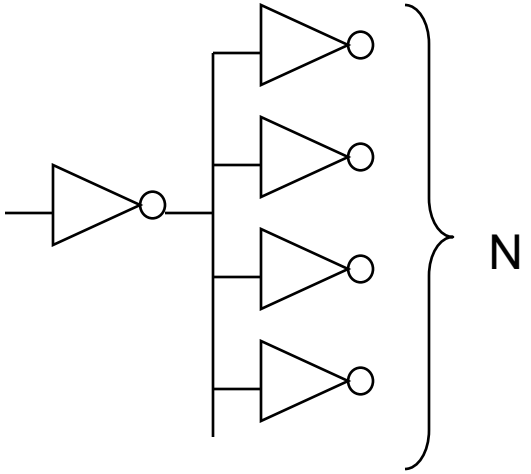
Vendor	Device	Technology [nm]	DSP	Memory [Mb]	TID (Krad)	SEL immunity (LET)
Microsemi	RTAX4000	150	0	0.53	300	$< 117 MeV.cm2/mg$
Microsemi	RTG4	65	462	3.76	160	$< 103 MeV.cm2/mg$
Microsemi	PolarFire RTPF500T	28	1481	33	100	$< 80 MeV.cm2/mg$
nanoeXplore	BRAVE Large	65	384	9.0	100	$< 62 MeV.cm2/mg$
nanoeXplore	BRAVE Ultra	28	1344	31.5	-	-
Xilinx	Virtex-5QV	65	320	10.7	100	$< 100 MeV.cm2/mg$
Xilinx	Kintex XQRKU060	20	2760	38.0	100	$< 79.2 MeV.cm2/mg$

Fundamental Design Metrics

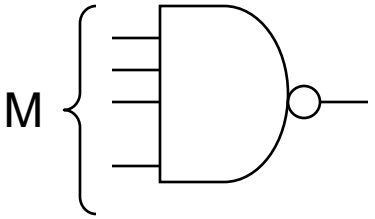
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Fan-In and Fan-Out

- Fan-out – number of load gates connected to the output of the driving gate
  - gates with large fan-out are slower



- Fan-in – the number of inputs to the gate
  - gates with large fan-in are bigger and slower



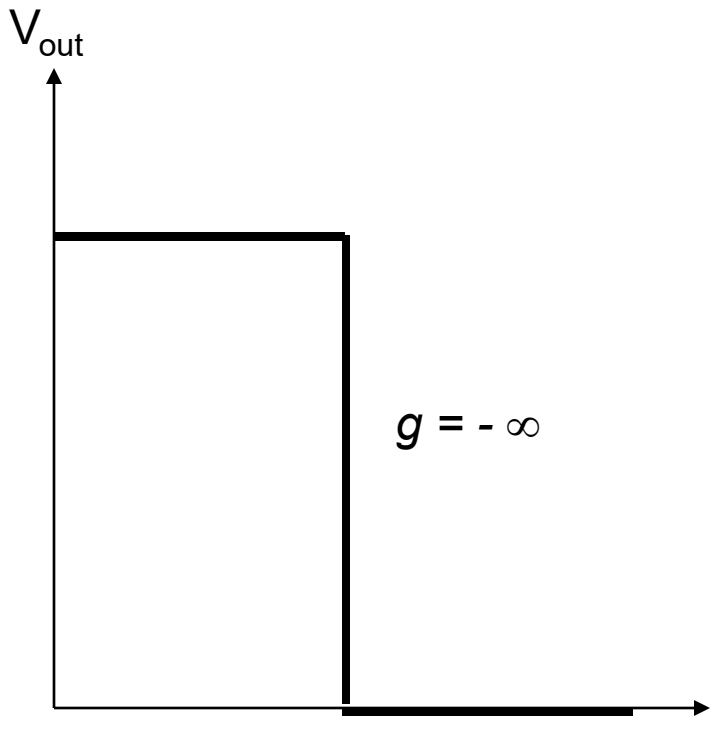
Directivity

- A gate must be *undirectional*: changes in an output level should not appear at any unchanging input of the same circuit
  - In real circuits *full* directivity is an illusion (e.g., due to capacitive coupling between inputs and outputs)

- Key metrics: *output impedance* of the driver and *input impedance* of the receiver
  - ideally, the output impedance of the driver should be zero and
  - input impedance of the receiver should be infinity

The Ideal Inverter

- The ideal gate should have
  - infinite gain in the transition region
  - a gate threshold located in the middle of the logic swing
  - high and low noise margins equal to half the swing
  - input and output impedances of infinity and zero, resp.



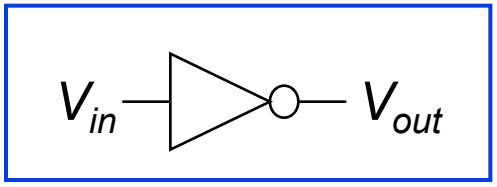
$R_i = \infty$

$R_o = 0$

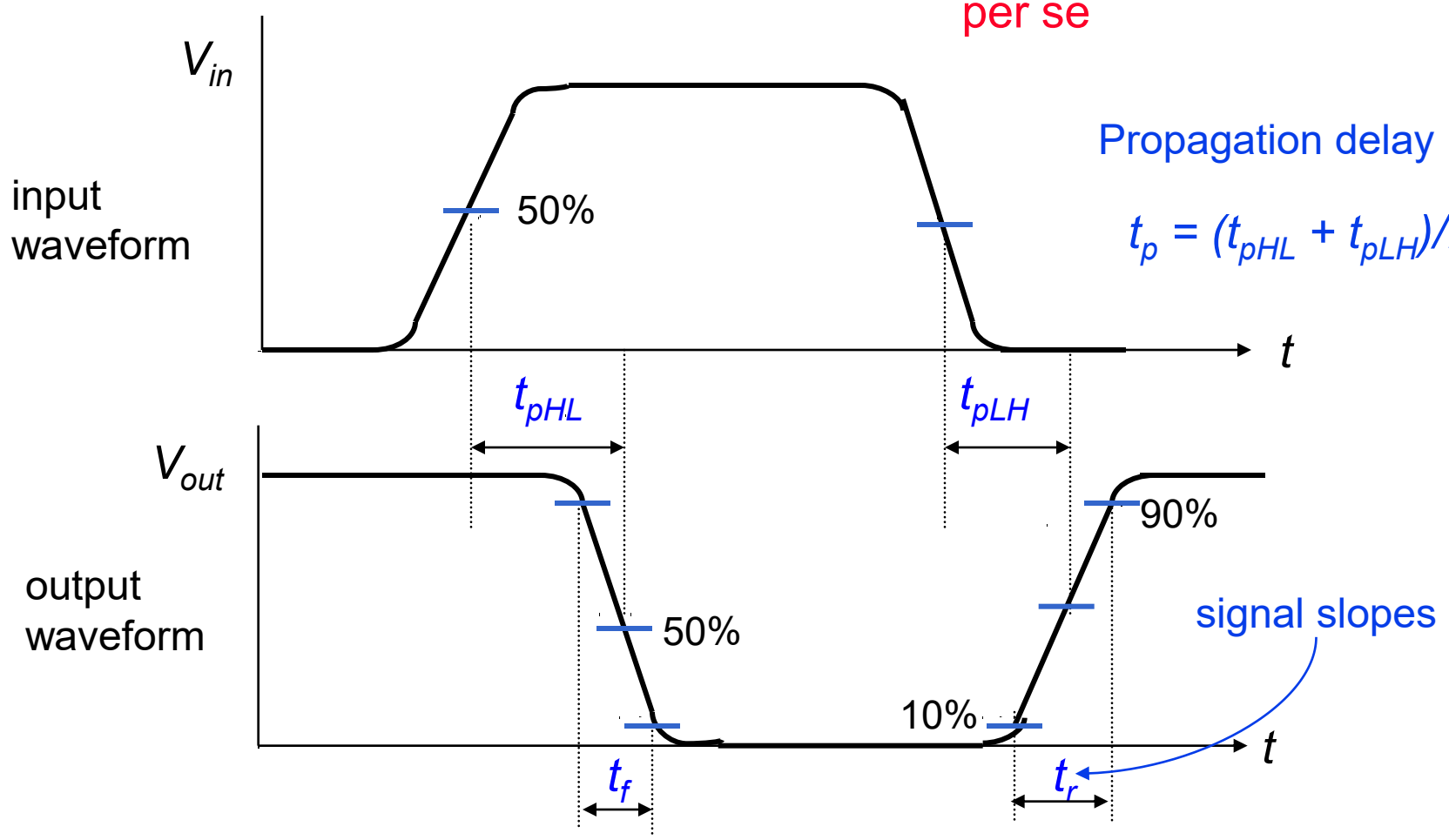
Fanout = ∞

$NM_H = NM_L = VDD/2$

Delay Definitions

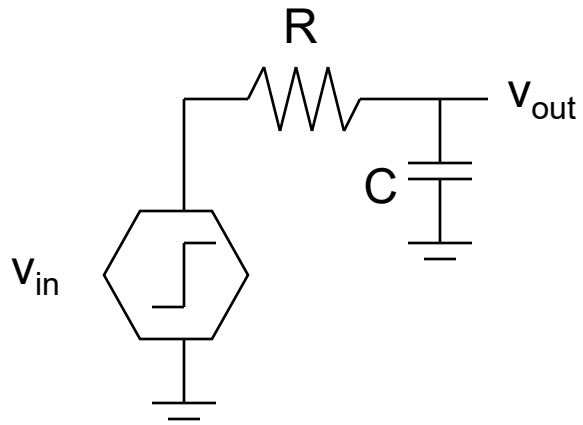


T_p is an artificial gate quality metric,no physical meaning per se



Modeling Propagation Delay

- Model circuit as first-order RC network



$$v_{out}(t) = (1 - e^{-t/\tau})V$$

where $\tau = RC$

Time to reach 50% point is
$t = \ln(2)\,\tau = 0.69\,\tau$

Time to reach 90% point is
$t = \ln(9)\,\tau = 2.2\,\tau$

- Matches the delay of an inverter gate

Power and Energy Dissipation

- Power consumption: how much energy is consumed per operation and how much heat the circuit dissipates
  - supply line sizing (determined by peak power)

$$P_{peak} = V_{dd} i_{peak}$$

  - battery lifetime (determined by average power dissipation)

$$p(t) = v(t)i(t) = V_{dd} i(t) \qquad P_{avg} = 1/T \int p(t)\,dt = V_{dd}/T \int i_{dd}(t)\,dt$$

  - packaging and cooling requirements

- Two important components: static and dynamic

$$P \text{ (watts)} = C_L V_{dd}^2 f_{0\to1} + t_{sc} V_{dd} I_{peak} f_{0\to1} + V_{dd} I_{leakage}$$

$$f_{0\to1} = P_{0\to1} * f_{clock}$$

Power and Energy Dissipation

- Propagation delay and the power consumption of a gate are related
- Propagation delay is (mostly) determined by the speed at which a given amount of energy can be stored on the gate capacitors
  - the faster the energy transfer (higher power dissipation) the faster the gate
- For a given technology and gate topology, the product of the power consumption and the propagation delay is a constant
  - Power-delay product (PDP) – energy consumed by the gate per switching event
- An ideal gate is one that is fast and consumes little energy, so the ultimate quality metric is
  - Energy-delay product (EDP) = power-delay ^{2}

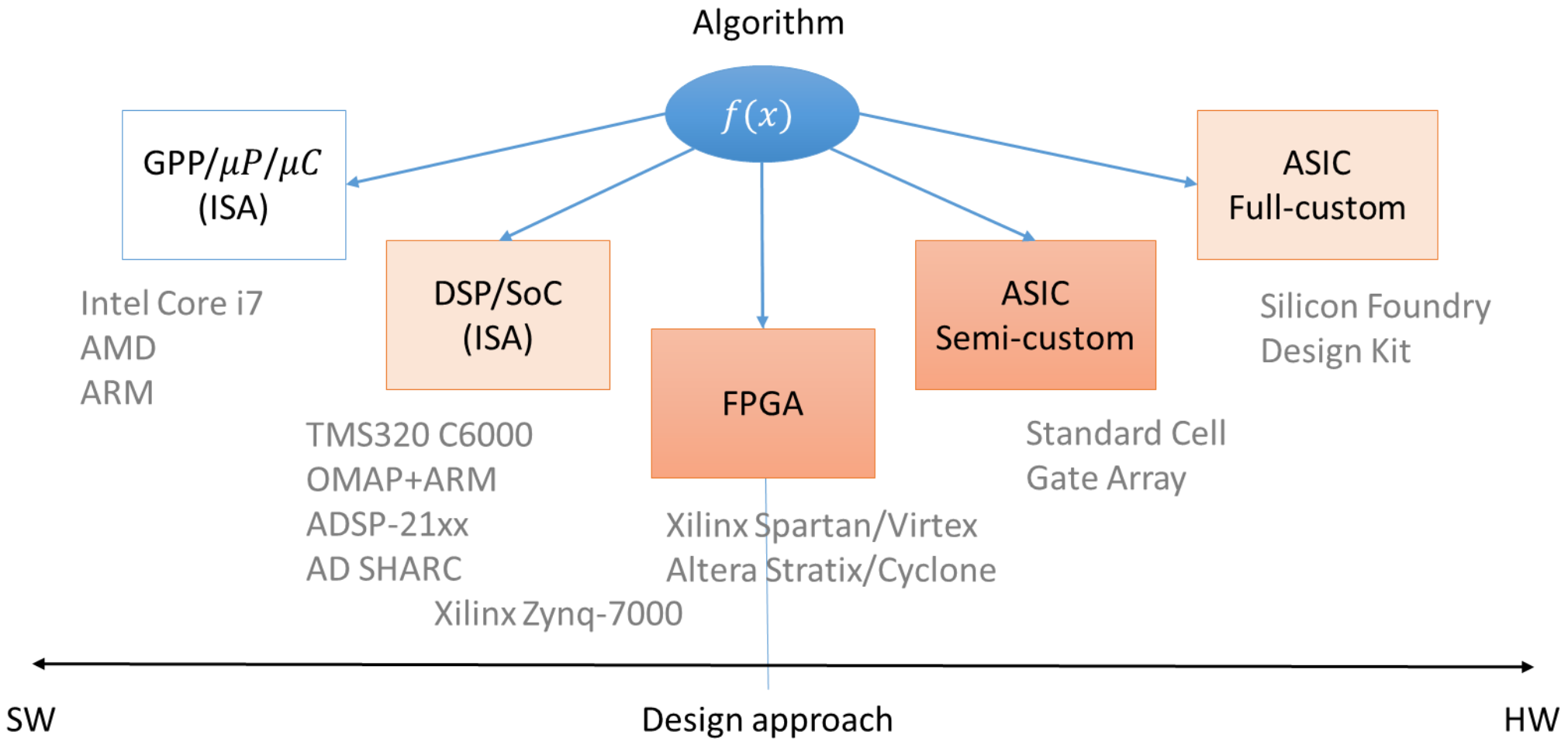
Summary

- Digital integrated circuits have come a long way and still have quite some potential left for the coming decades
- Some interesting challenges ahead
  - Getting a clear perspective on the challenges and potential solutions is the purpose of this course
- Understanding the design metrics that govern digital design is crucial
  - Cost, reliability, speed, power and energy dissipation

Outline

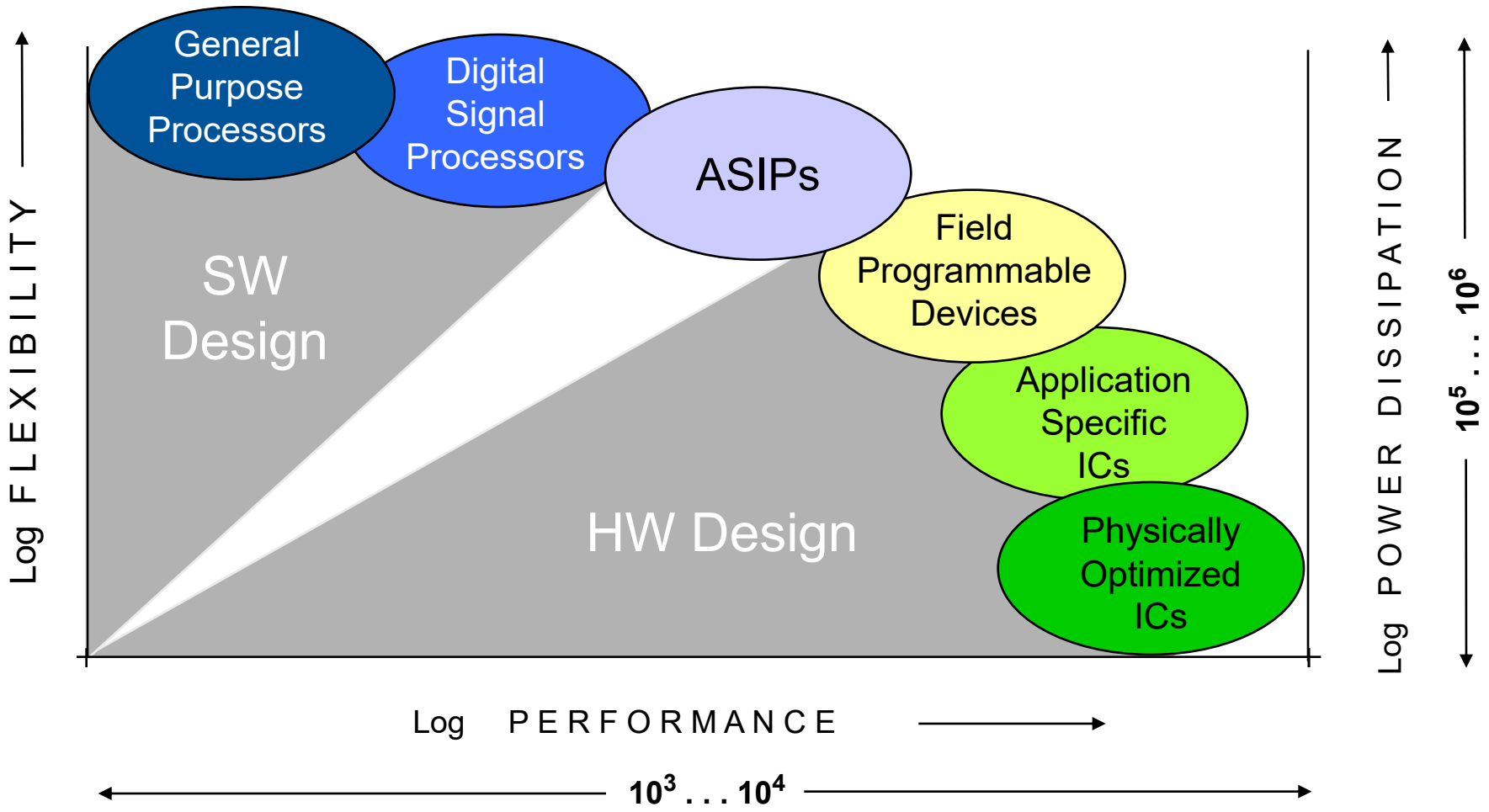
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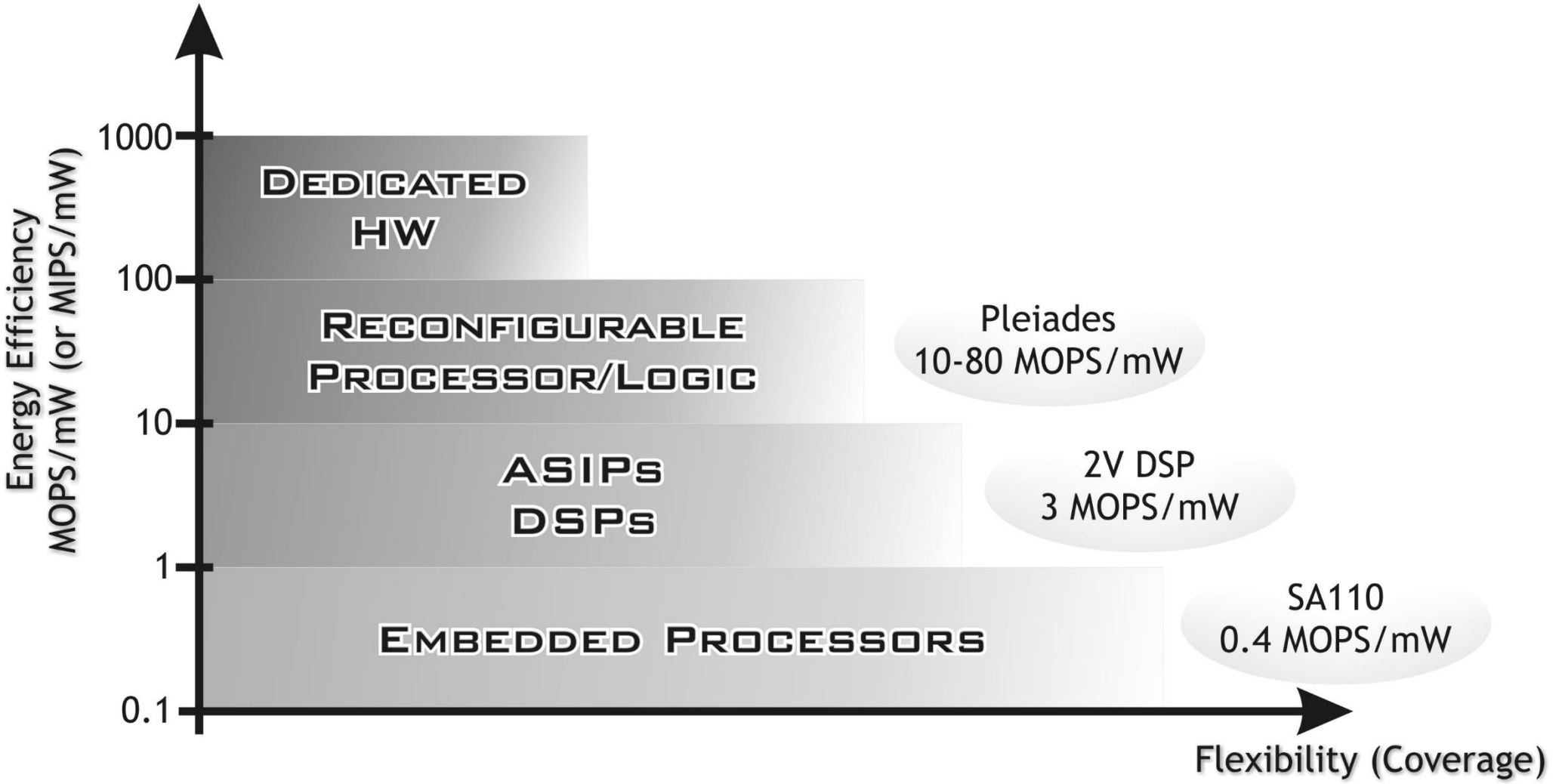
Design is the art of managing engineering trade-offs !

The Energy-Flexibility Gap



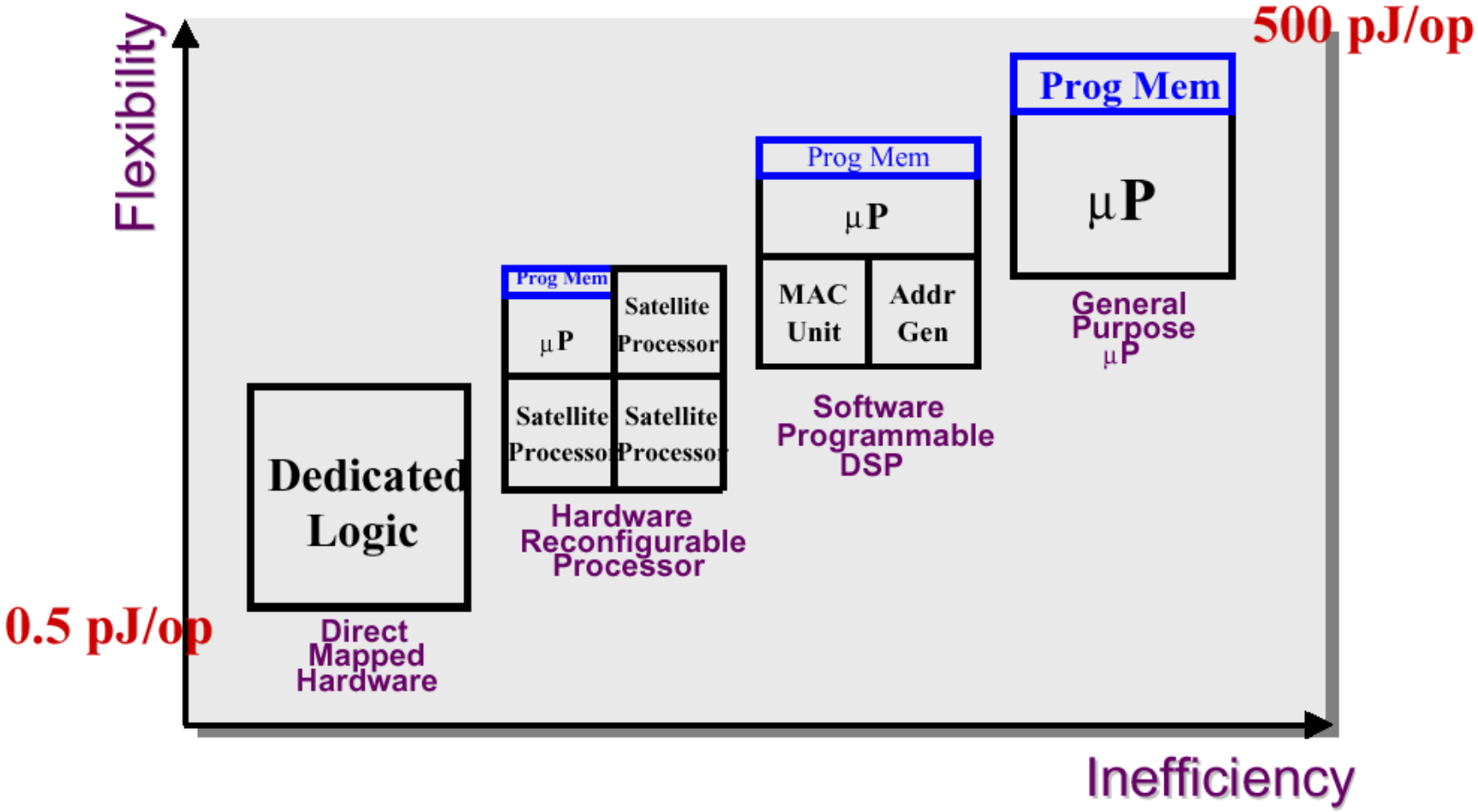
Source: T.Noll, RWTH Aachen

The Energy-Flexibility Gap



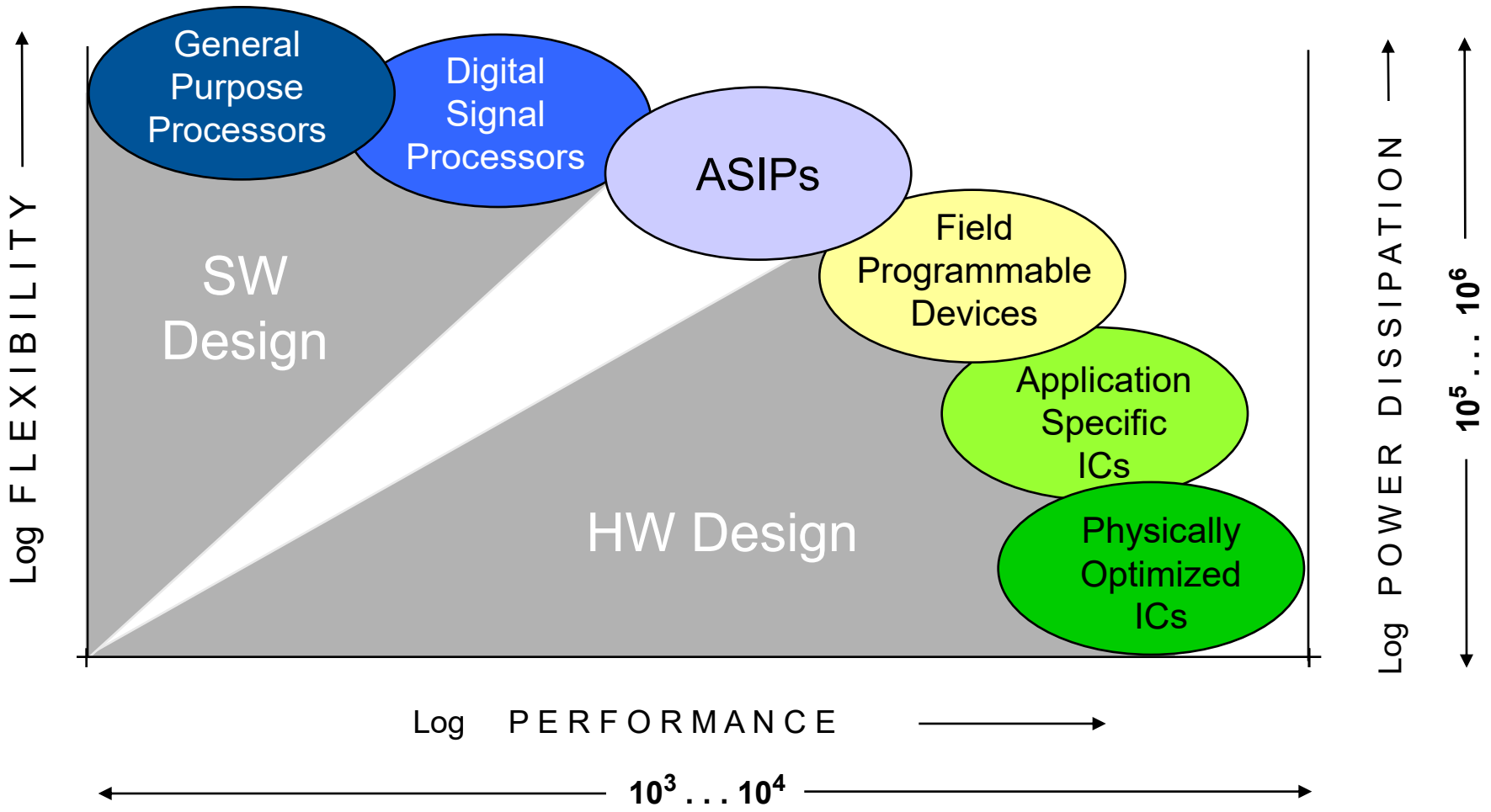
Source: Rabaey, BRWC

The Energy-Flexibility Gap



Source: Rabaey, BRWC

The Energy-Flexibility Gap



Source: T.Noll, RWTH Aachen

SoC elements

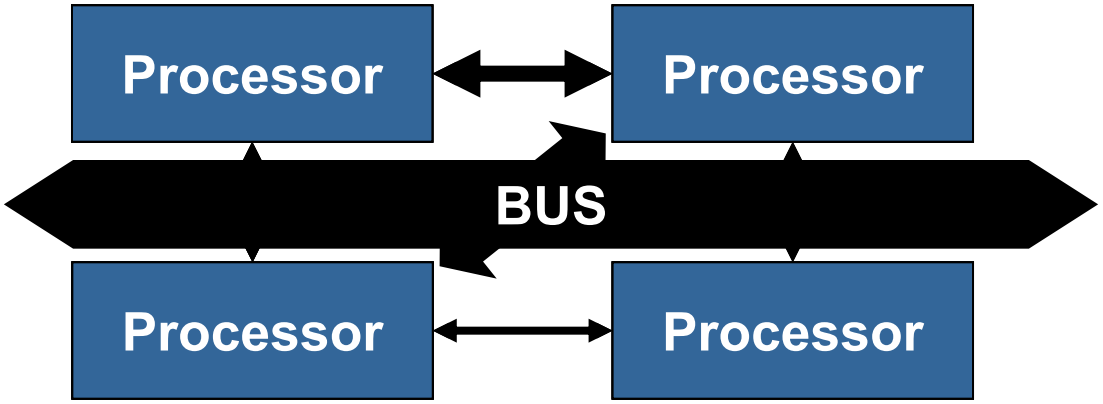
programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	

How to address growing SoC complexity?

SoC elements

programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	

MP-SoC solution:

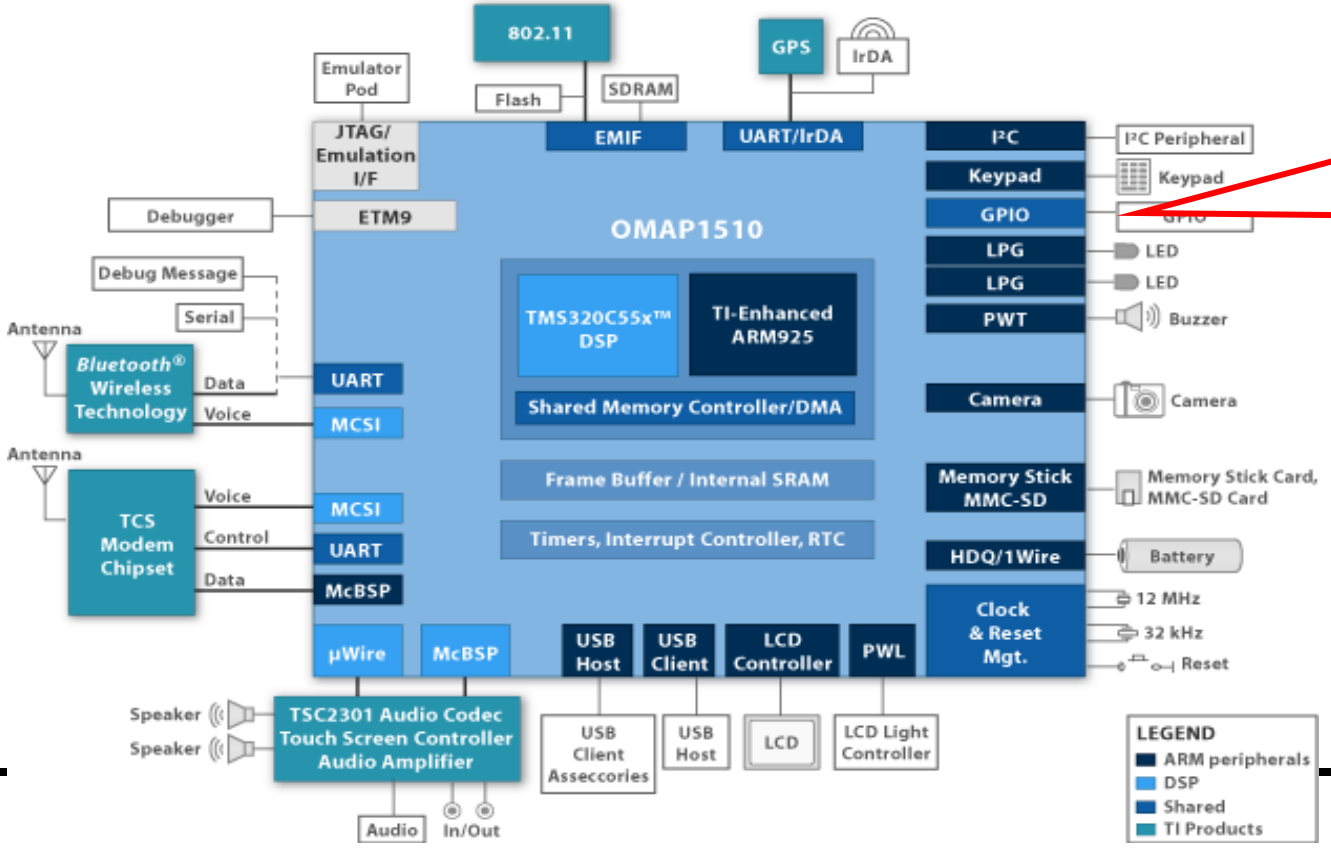


\+ Programmable solution
\+ Utilizing existing tools

\- Scheduling and Mapping may be difficult
\- Throughput constraints not achievable

SoC elements

programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	

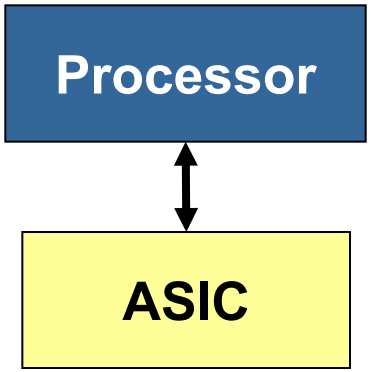


Commercial Example: TI OMAP (1510)

SoC elements

programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	

Programmable Solution and Accelerator:



\+ Programmable solution

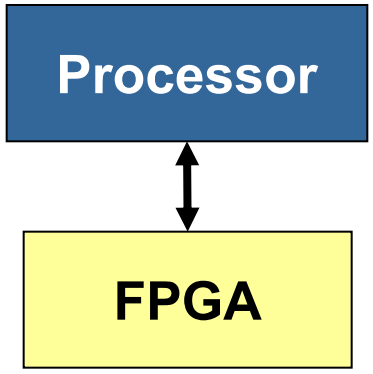
• Utilizing existing tools ?

\- Flexibility bound to given limitations:
  - Instruction Set
  - Processor/ASIC Interface
  - Memory Interface
  - Architecture Timing

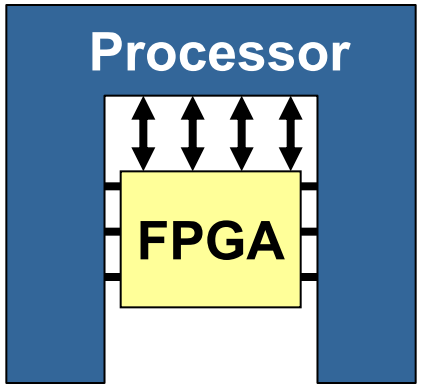
SoC elements

programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	

Programmable Solution and Accelerator:



loosely coupled



tightly coupled
(embedded into infrastructure)

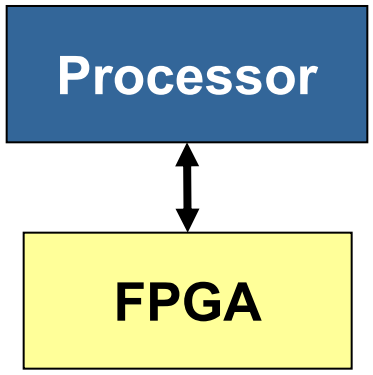
\+ Programmable solution
\+ High Performance

• Utilizing existing tools ?

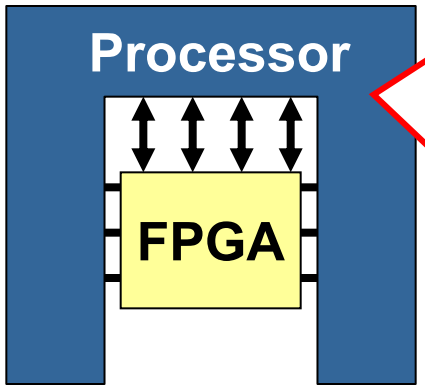
\- Flexibility bound to given limitations:
  - Instruction Set
  - Processor/FPGA Interface
  - Memory Interface
  - Architecture Timing

SoC elements

programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	



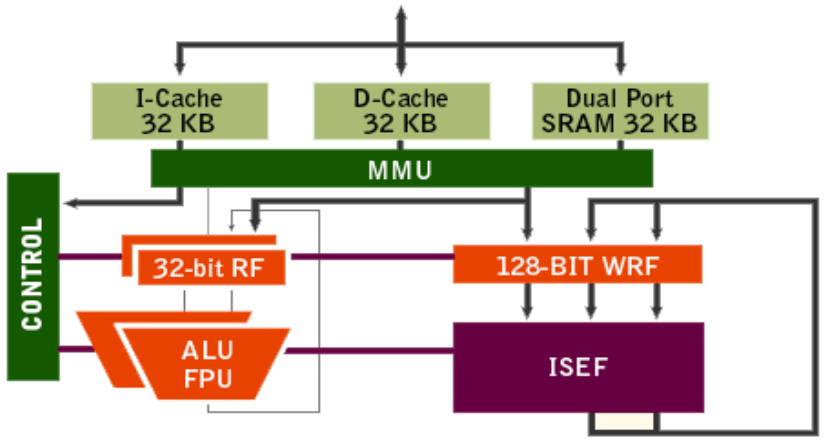
loosely coupled



tightly coupled
(embedded into infrastructure)

Comercial Example: Stretch Inc.

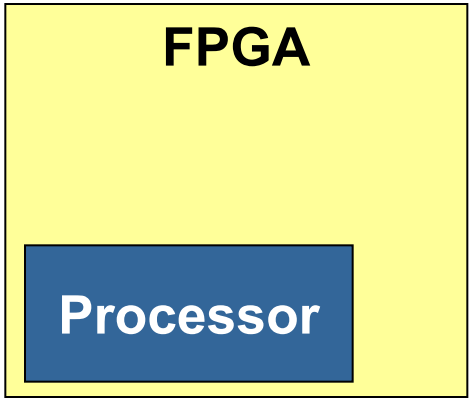
load/store up to 3x/2x 128 bit
User-defined extensions to the core ISA



SoC elements

programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	

Programmable Solution and Accelerator:



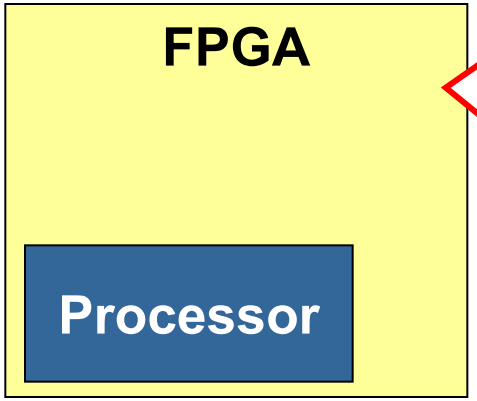
\+ Programmable solution

• Utilizing existing tools ?

\- Energy Efficiency
\- Area/Timing Efficency
\- Flexibility bound to given limitations:
  - Instruction Set
  - Processor/FPGA Interface
  - Memory Interface
  - Architecture Timing

SoC elements

programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	



Comercial Example:

XILINX, Virtex-II Pro

Single or multiple IBM PowerPC Processors are embedded in the FPGA (Platform FPGA).

XILINX, Virtex-4 FPGA family

Logic Cells, RAM, XtremeDSP™ Slices, PowerPC Processor Blocks,
10/100/1000 Ethernet MAC Blocks, RocketIO™ Serial Transceivers

SoC elements

programmable	reconfigurable	fixed
DSP, µC, RISC, GPP, …	FPGA / eFPGA	ASIC
Compiler, Assembler, Linker, Simulator, Debugger	Gate-Level Synthesis Tools, Power Analysis Tools, HW-Simulator	

To map the Architecture to the Application means:

>> Consider jointly HW and SW for an efficient solution

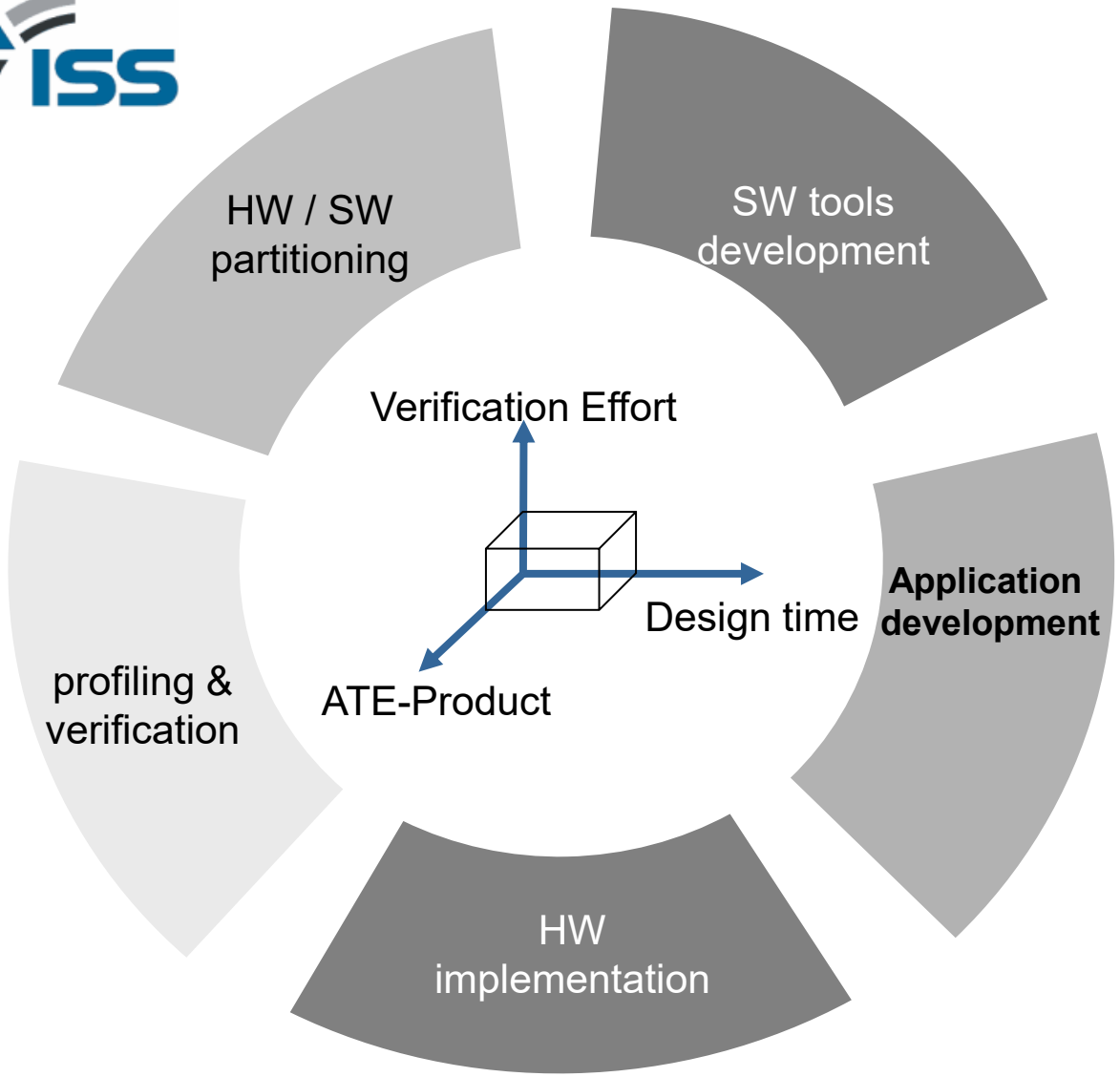
>> Identify hotspots of the application and ...

>> implement complex instructions.

Without being limited to given structures or interfaces!

Application Specific Instruction Set Processor (ASIP)

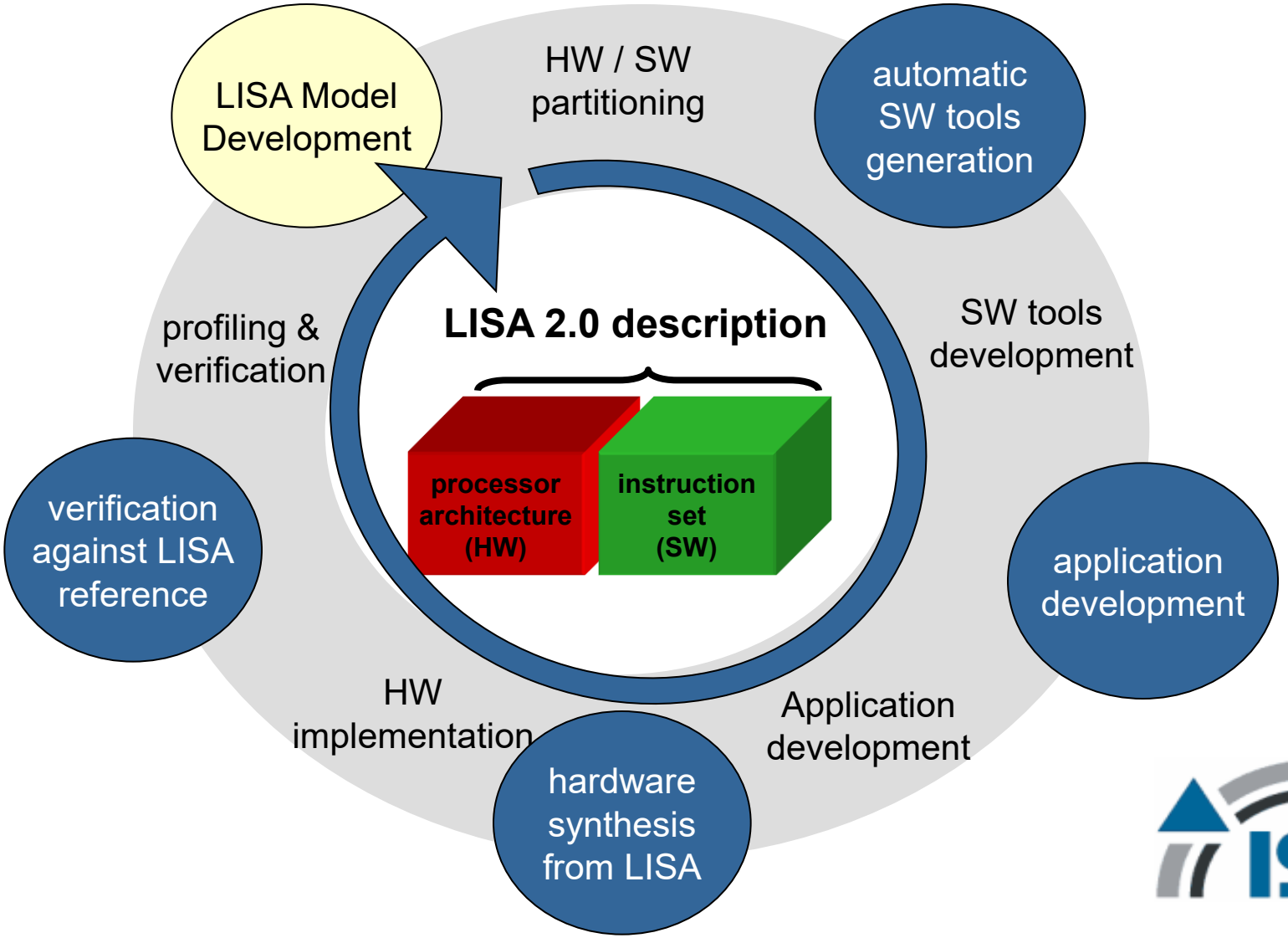
How to move through the design space?



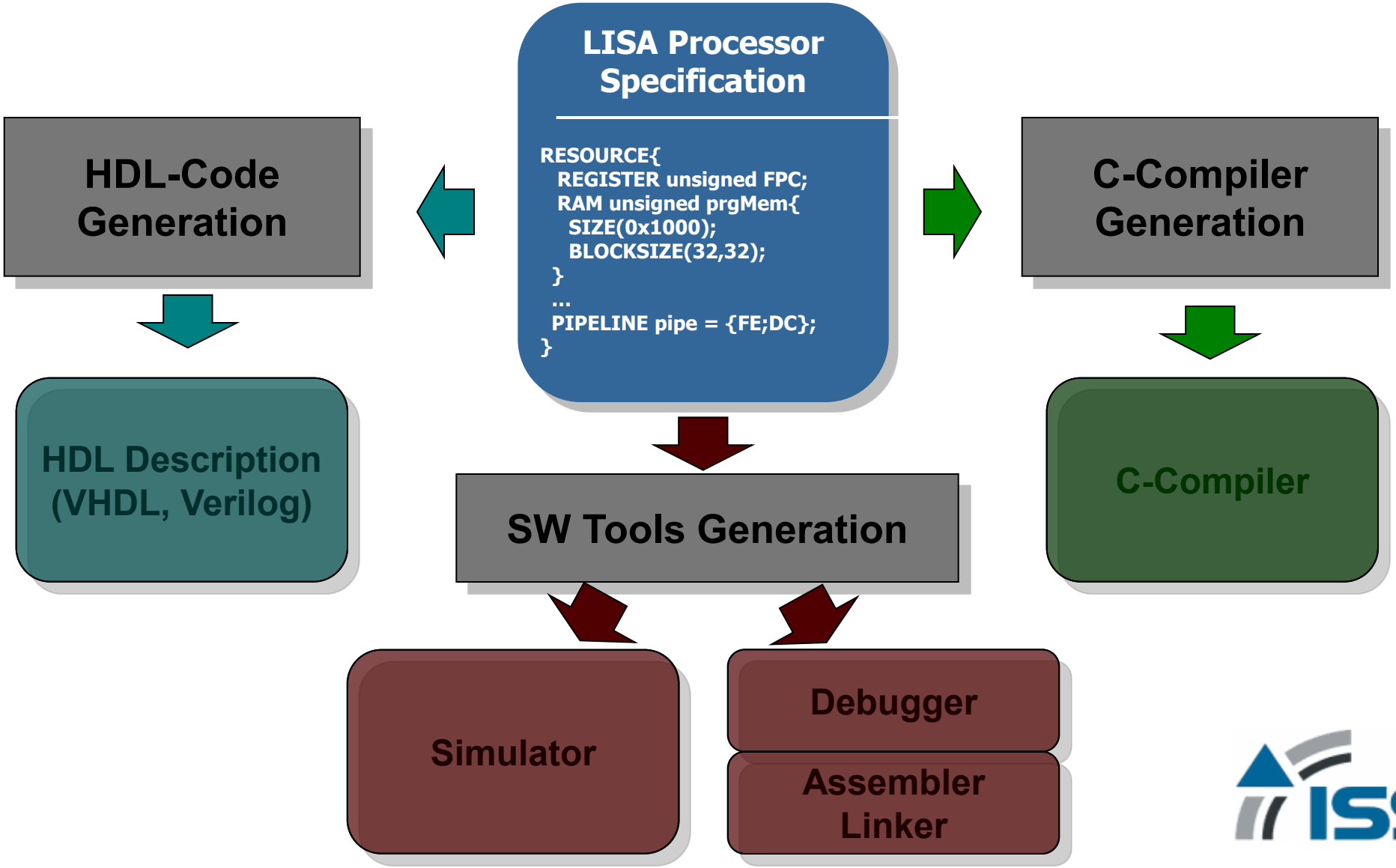
Trade-offs:

- Area vs. Timing vs. Energy
- Design Time vs. ATE-Product
- Architectural Features vs. Verification Effort
- Verification Effort vs. ATE-Product
- Flexibility vs. ATE-Product

ADL (Architecture Design Language) LISA

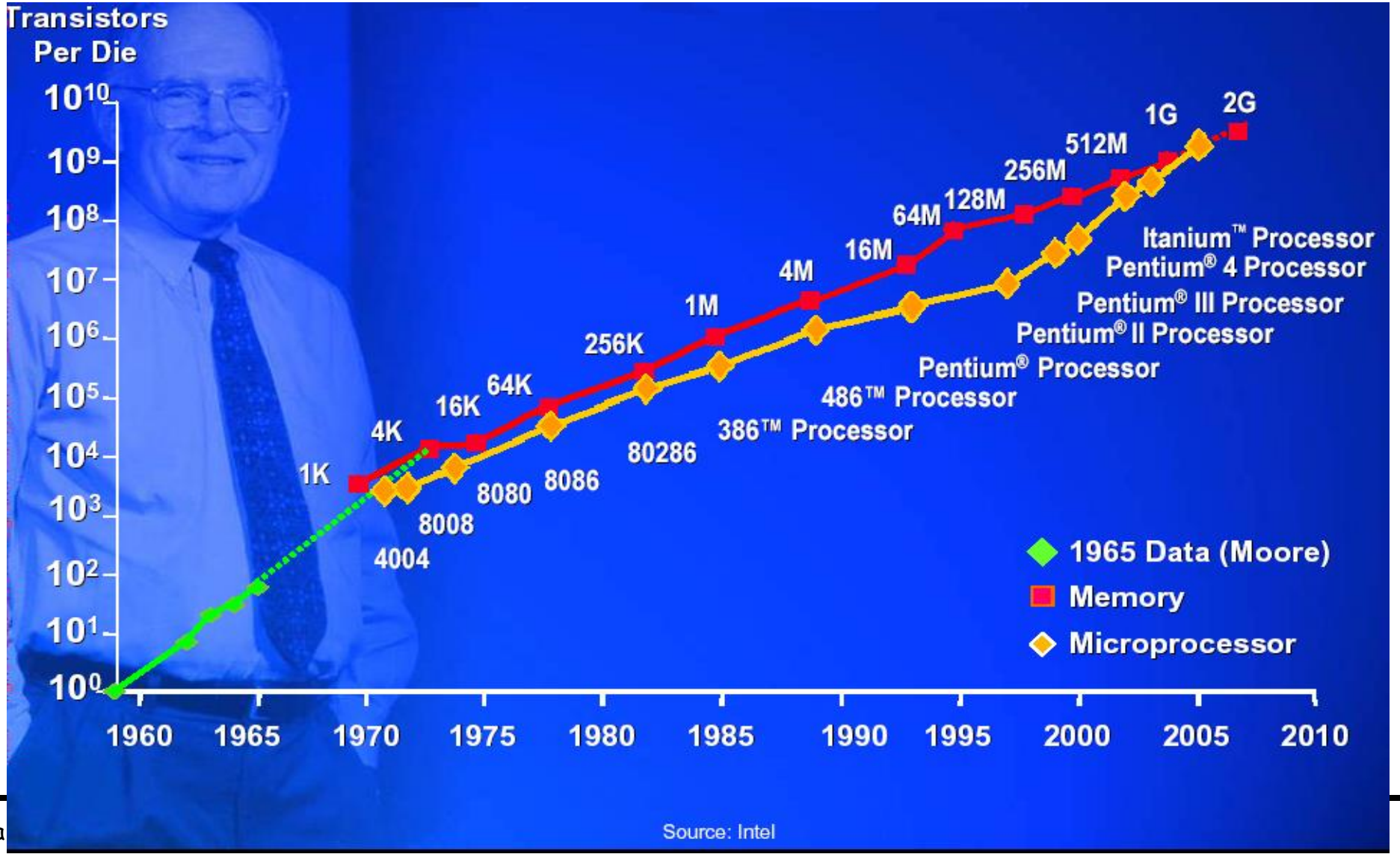


Centralized Processor Design with LISA



Moore's Law

- Original "law": number of components on IC doubles every year
- Later modified to doubling every 18 months



Moore's Law

Cramming More Components onto Integrated Circuits

GORDON E. MOORE, LIFE FELLOW, IEEE

With unit cost falling as the number of components per circuit rises, by 1975 economics may dictate squeezing as many as 65 000 components on a single silicon chip.

The future of integrated electronics is the future of electronics itself. The advantages of integration will bring about a proliferation of electronics, pushing this science into many new areas.

Integrated circuits will lead to such wonders as home computers—or at least terminals connected to a central computer—automatic controls for automobiles, and personal portable communications equipment. The electronic wristwatch needs only a display to be feasible today.

But the biggest potential lies in the production of large systems. In telephone communications, integrated circuits in digital filters will separate channels on multiplex equipment. Integrated circuits will also switch telephone circuits and perform data processing.

Computers will be more powerful, and will be organized in completely different ways. For example, memories built of integrated electronics may be distributed throughout the machine instead of being concentrated in a central unit. In addition, the improved reliability made possible by integrated circuits will allow the construction of larger processing units. Machines similar to those in existence today will be built at lower costs and with faster turnaround.

I. Present and Future

By integrated electronics, I mean all the various technologies which are referred to as microelectronics today as well as any additional ones that result in electronics functions supplied to the user as irreducible units. These technologies were first investigated in the late 1950's. The object was to miniaturize electronics equipment to include increasingly complex electronic functions in limited space with minimum weight. Several approaches evolved, including microassembly techniques for individual components, thin-film structures, and semiconductor integrated circuits.

Reprinted from Gordon E. Moore, "Cramming More Components onto Integrated Circuits," *Electronics*, pp. 114–117, April 19, 1965.
Publisher Item Identifier S 0018-9219(98)00753-1.

Each approach evolved rapidly and converged so that each borrowed techniques from another. Many researchers believe the way of the future to be a combination of the various approaches.

The advocates of semiconductor integrated circuitry are already using the improved characteristics of thin-film resistors by applying such films directly to an active semiconductor substrate. Those advocating a technology based upon films are developing sophisticated techniques for the attachment of active semiconductor devices to the passive film arrays.

Both approaches have worked well and are being used in equipment today.

II. The Establishment

Integrated electronics is established today. Its techniques are almost mandatory for new military systems, since the reliability, size, and weight required by some of them is achievable only with integration. Such programs as Apollo, for manned moon flight, have demonstrated the reliability of integrated electronics by showing that complete circuit functions are as free from failure as the best individual transistors.

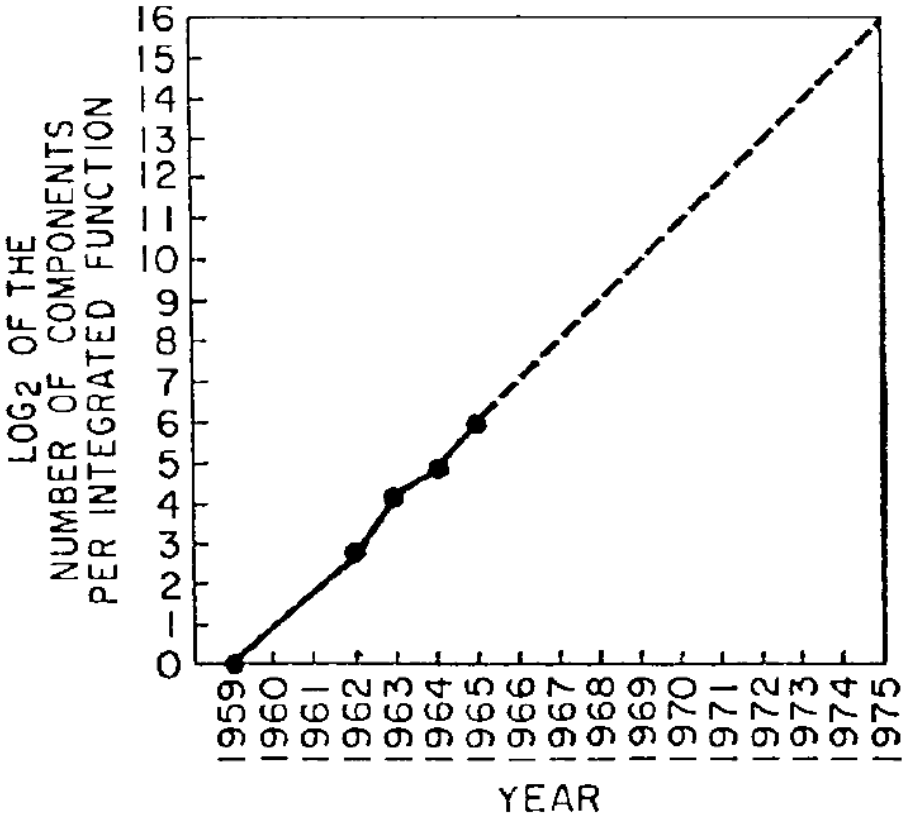
Most companies in the commercial computer field have machines in design or in early production employing integrated electronics. These machines cost less and perform better than those which use "conventional" electronics.

Instruments of various sorts, especially the rapidly increasing numbers employing digital techniques, are starting to use integration because it cuts costs of both manufacture and design.

The use of linear integrated circuitry is still restricted primarily to the military. Such integrated functions are expensive and not available in the variety required to satisfy a major fraction of linear electronics. But the first applications are beginning to appear in commercial electronics, particularly in equipment which needs low-frequency amplifiers of small size.

III. Reliability Counts

In almost every case, integrated electronics has demonstrated high reliability. Even at the present level of pro-



Dennard's Scaling

Design of Ion-Implanted MOSFET's with Very Small Physical Dimensions

ROBERT H. DENNARD, MEMBER, IEEE, FRITZ H. GAENSSLEN, HWA-NIEN YU, MEMBER, IEEE, V. LEO RIDEOUT, MEMBER, IEEE, ERNEST BASSOUS, AND ANDRE R. LEBLANC, MEMBER, IEEE

Classic Paper

This paper considers the design, fabrication, and characterization of very small MOSFET switching devices suitable for digital integrated circuits using dimensions of the order of 1 μ. Scaling relationships are presented which show how a conventional MOSFET can be reduced in size. An improved small device structure is presented that uses ion implantation to provide shallow source and drain regions and a nonuniform substrate doping profile. One-dimensional models are used to predict the substrate doping profile and the corresponding threshold voltage versus source voltage characteristic. A two-dimensional current transport model is used to predict the relative degree of short-channel effects for different device parameter combinations. Polysilicon-gate MOSFET's with channel lengths as short as 0.5 μ were fabricated, and the device characteristics measured and compared with predicted values. The performance improvement expected from using these very small devices in highly miniaturized integrated circuits is projected.

I. List of Symbols

α	Inverse semilogarithmic slope of subthreshold characteristic.
D	Width of idealized step function profile for channel implant.
ΔW_f	Work function difference between gate and substrate.
$\epsilon_{Si}, \epsilon_{ox}$	Dielectric constants for silicon and silicon dioxide.
I_d	Drain current.
k	Boltzmann's constant.
κ	Unitless scaling constant.
L	MOSFET channel length.
μ_{eff}	Effective surface mobility.
n_i	Intrinsic carrier concentration.
N_a	Substrate acceptor concentration.
Ψ_s	Band bending in silicon at the onset of strong inversion for zero substrate voltage.
Ψ_b	Built-in junction potential.
q	Charge on the electron.
Q_{eff}	Effective oxide charge.
t_{ox}	Gate oxide thickness.
T	Absolute temperature.
V_d, V_s, V_g, V_{sub}	Drain, source, gate and substrate voltages.
V_{ds}	Drain voltage relative to source.
V_{s-sub}	Source voltage relative to sustrate.
V_t	Gate threshold voltage.
w_s, w_d	Source and drain depletion layer widths.
W	MOSFET channel width.

This paper is reprinted from IEEE Journal of Solid-State Circuits, vol. SC-9, no. 5, pp. 256–268, October 1974.
Publisher Item Identifier S 0018-9219(99)02196-9.

II. Introduction

New high resolution lithographic techniques for forming semiconductor integrated circuit patterns offer a decrease in linewidth of five to ten times over the optical contact masking approach which is commonly used in the semiconductor industry today. Of the new techniques, electron beam pattern writing has been widely used for experimental device fabrication [1]–[4] while X-ray lithography [5] and optical projection printing [6] have also exhibited high-resolution capability. Full realization of the benefits of these new high-resolution lithographic techniques requires the development of new device designs, technologies, and structures which can be optimized for very small dimensions.

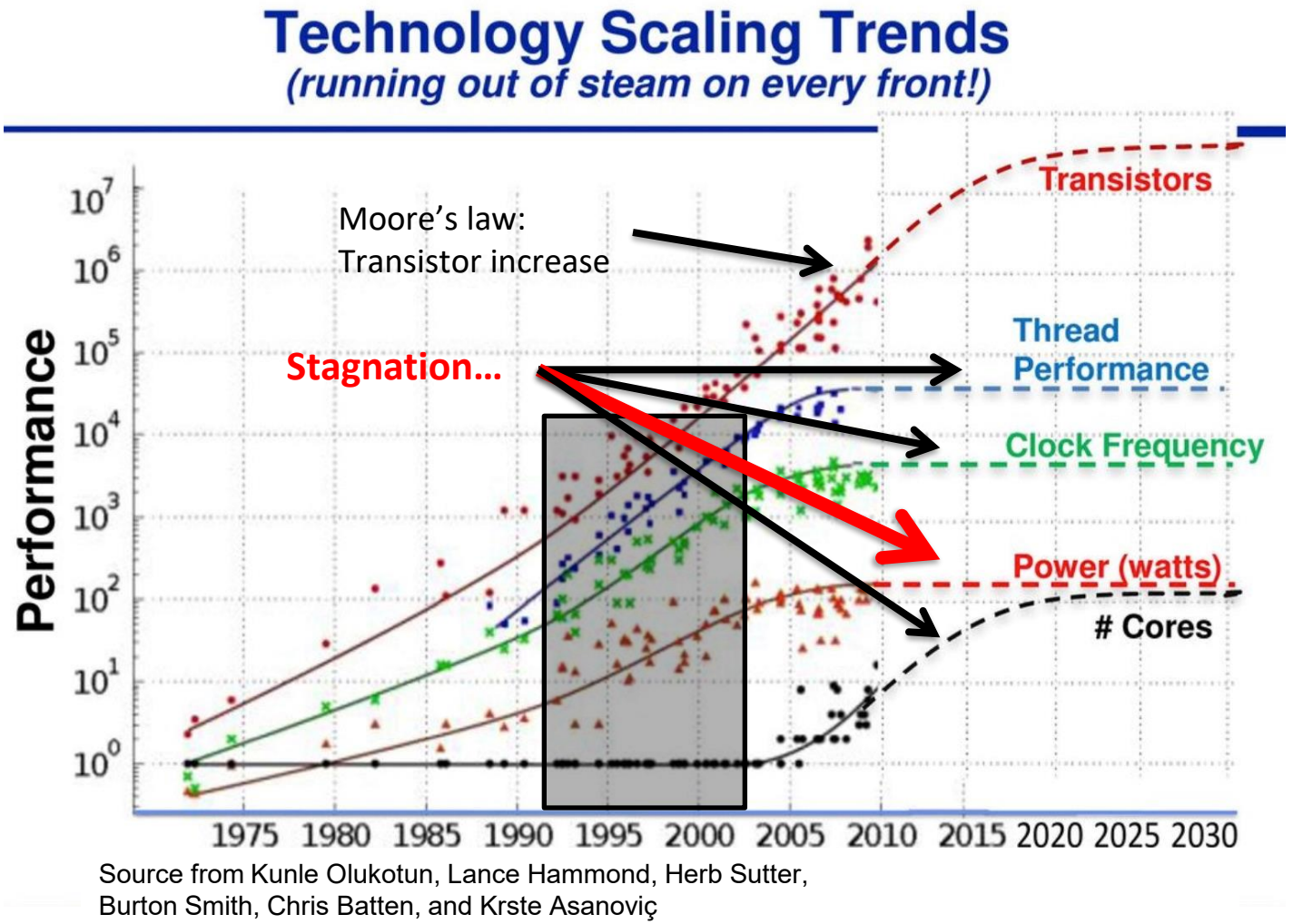
This paper concerns the design, fabrication, and characterization of very small MOSFET switching devices suitable for digital integrated circuits using dimensions of the order of 1 μ. It is known that reducing the source-to-drain spacing (i.e., the channel length) of an FET leads to undesirable changes in the device characteristics. These changes become significant when the depletion regions surrounding the source and drain extend over a large portion of the region in the silicon substrate under the gate electrode. For switching applications, the most undesirable "short-channel" effect is a reduction in the gate threshold voltage at which the device turns on, which is aggravated

0018–9219/99$10.00 © 1999 IEEE

Dennard scaling, states, roughly, that as transistors get smaller their power density stays constant, so that the power use stays in proportion with area: both voltage and current scale (downward) with length.

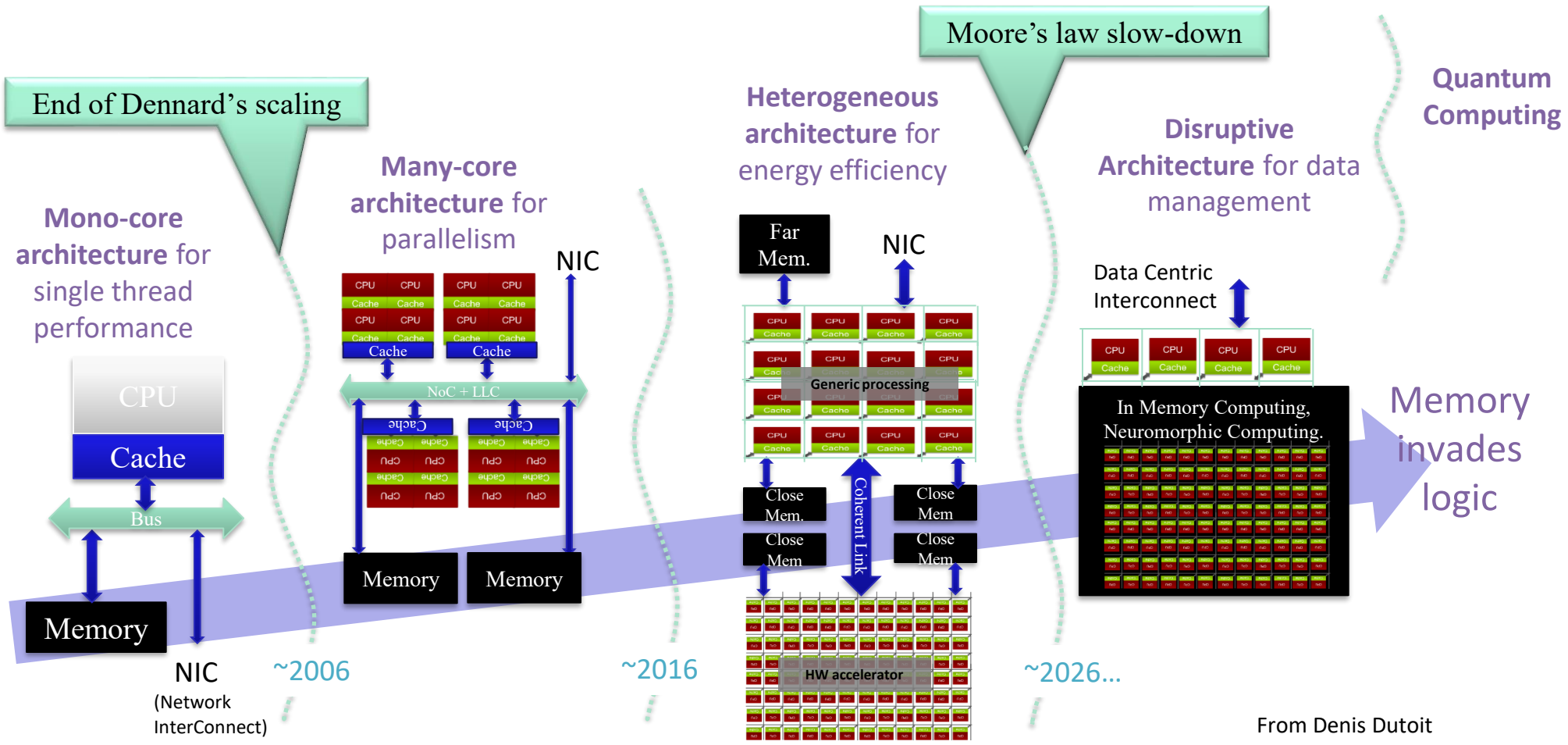
The breakdown of Dennard scaling and resulting inability to increase clock frequencies significantly has caused most CPU manufacturers to focus on multicore processors as an alternative way to improve performance.

Moore 's law and Dennard scaling



Source from Kunle Olukotun, Lance Hammond, Herb Sutter, Burton Smith, Chris Batten, and Krste Asanoviç

Processor architecture evolution



From Denis Dutoit

Silicon Roadmap

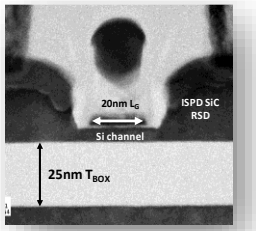
Silicon Quantum bits

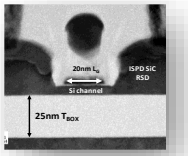
FDSOI

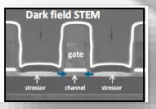
22FD

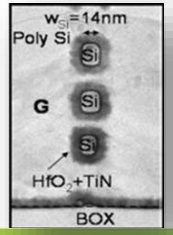
12FD

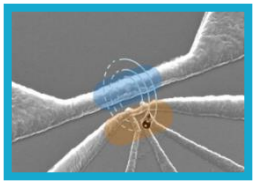
Next Gen











Non planar / trigate / stacked Nanowires

FinFET

28nm

14nm

10nm

2017

2019

7nm

5nm

Disruptive scaling

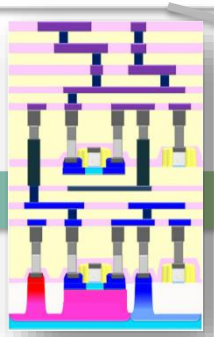
Hybrid logic

Steep slope devices

Mechanical switches

Si Quantum bits

Alternative to scaling and diversification

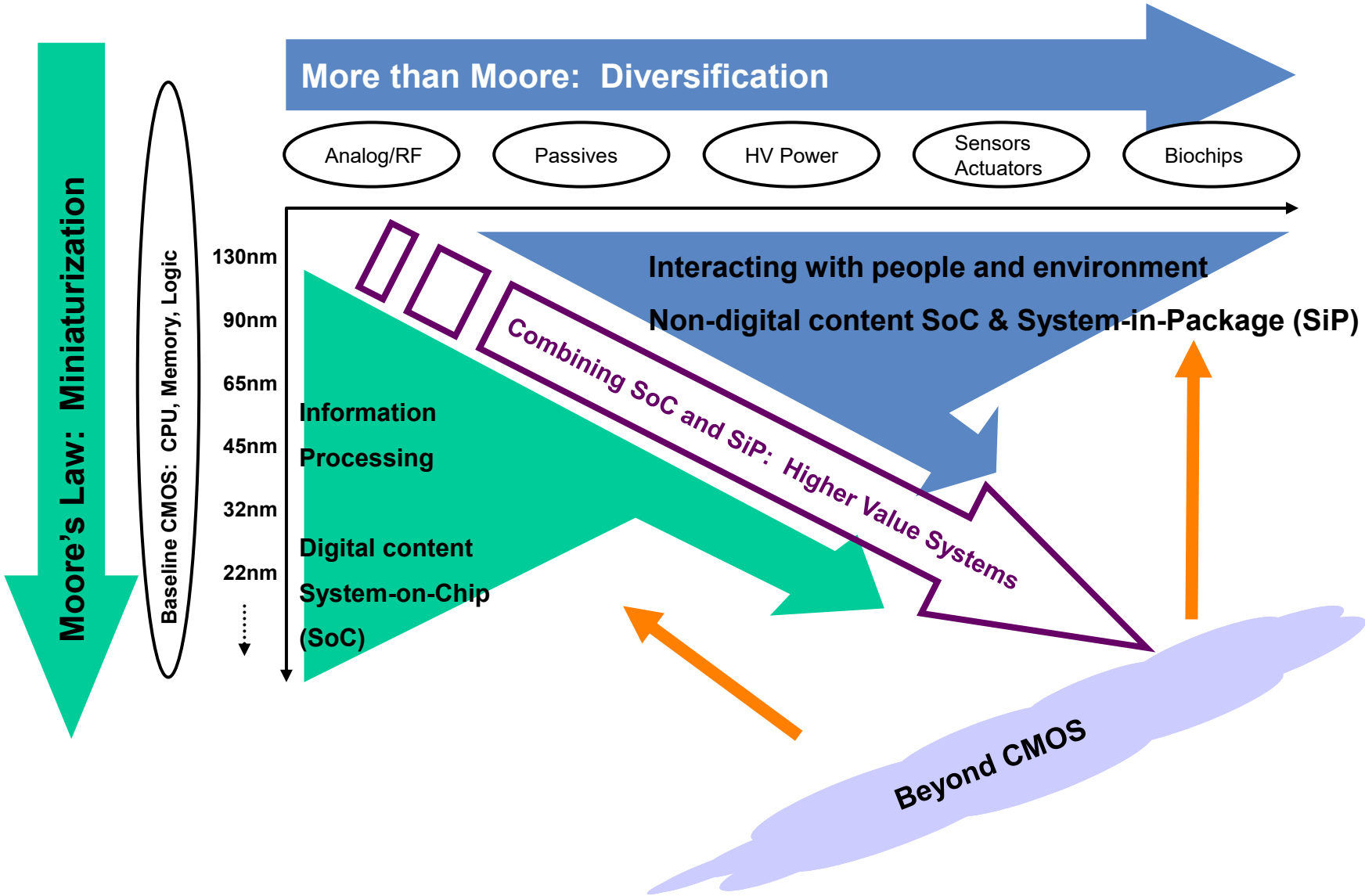


Monolithic 3D for 3D VLSI

Silicon Roadmap

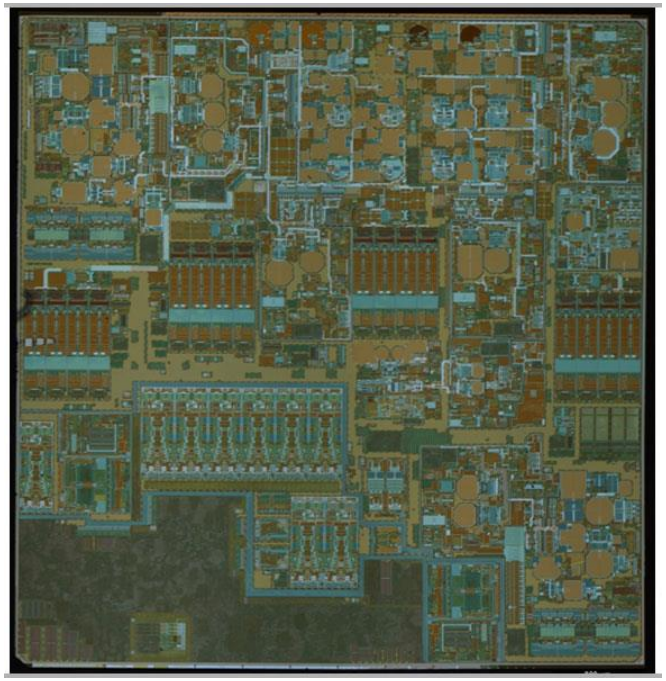
Nominal node		28nm	22nm	20nm	18nm	16nm	14nm	12nm	10nm	7nm	5nm (ITRS)
Intel	Lg		24				20		16	~12nm	10
	Fin Pitch		60 FinFET				FinFET 42		FinFET 34	FinFET	12
	CPP		90				70		54		32
	M1		80				52		36		16
	SRAM		HD 0.092µm2				0.0588µm2		0.0312µm2	0.027µm2	
	Year Publication		VLSI 2012				IEDM 2014		IEDM 2017/ISSCC2018	IEDM 2016	
	Risk Prod		2011				2014		1Q18		
Samsung	Lg	32		25	25		30		~20	~16	
	Fin Pitch	BULK		BULK	FDSOI		48 FinFET		Single Fin 42	Dual thin EUV 27	
	CPP	114		86	86		78		68	54/57	
	M1	90		64	64		64		51	36	
	SRAM	0.152µm2		0.084µm2			0.064/0.08µm2		0.04µm2	HD 6T SRAM 0.026µm2	
	Year Publication	ICSIST 2011		VLSI 2012			JSSC 2014		ISSCC/VLSI 2017	VLSI 2017/ISSCC2017-2018	
	Risk Prod	2011		2013			4Q-2015		1Q2017	2H-18	
TSMC	Lg	30	30	30		33		*25*	~20	~16	
	Fin Pitch	BULK	BULK	BULK		FinFET 45		*FinFET 45*	FinFET	FinFET 4th	
	CPP	118	105	90		90/80		*90/80*	64	57	
	M1	90	80	64		64		*64*	42	40	
	SRAM	0.155µm2	0.155µm2			0.07µm2			0.03µm2	0.027µm2	
	Year Publication	VLSI 2012	VLSI 2012	VLSI 2014		IEDM 2013		*6Track*	VLSI 2016	IEDM 2016	
	Risk Prod	2011	2018	2013		4Q-2015		*3Q2016*	4Q2016	3Q-17	
GF	Lg		28				30				
	Fin Pitch		FDSOI				48 Fin FET				
	CPP		90				78				
	M1		78				67				
	SRAM		0.110µm2				0.110µm2				
	Year Publication		IEDM 2016				IEDM 2016			C.Reita, C.Fenouillet-Beranger - CEA-LETI - 2018	
	Risk Prod		2016				2H-2016				

More Moore and More than Moore domains



Systems on Chip (SoC)

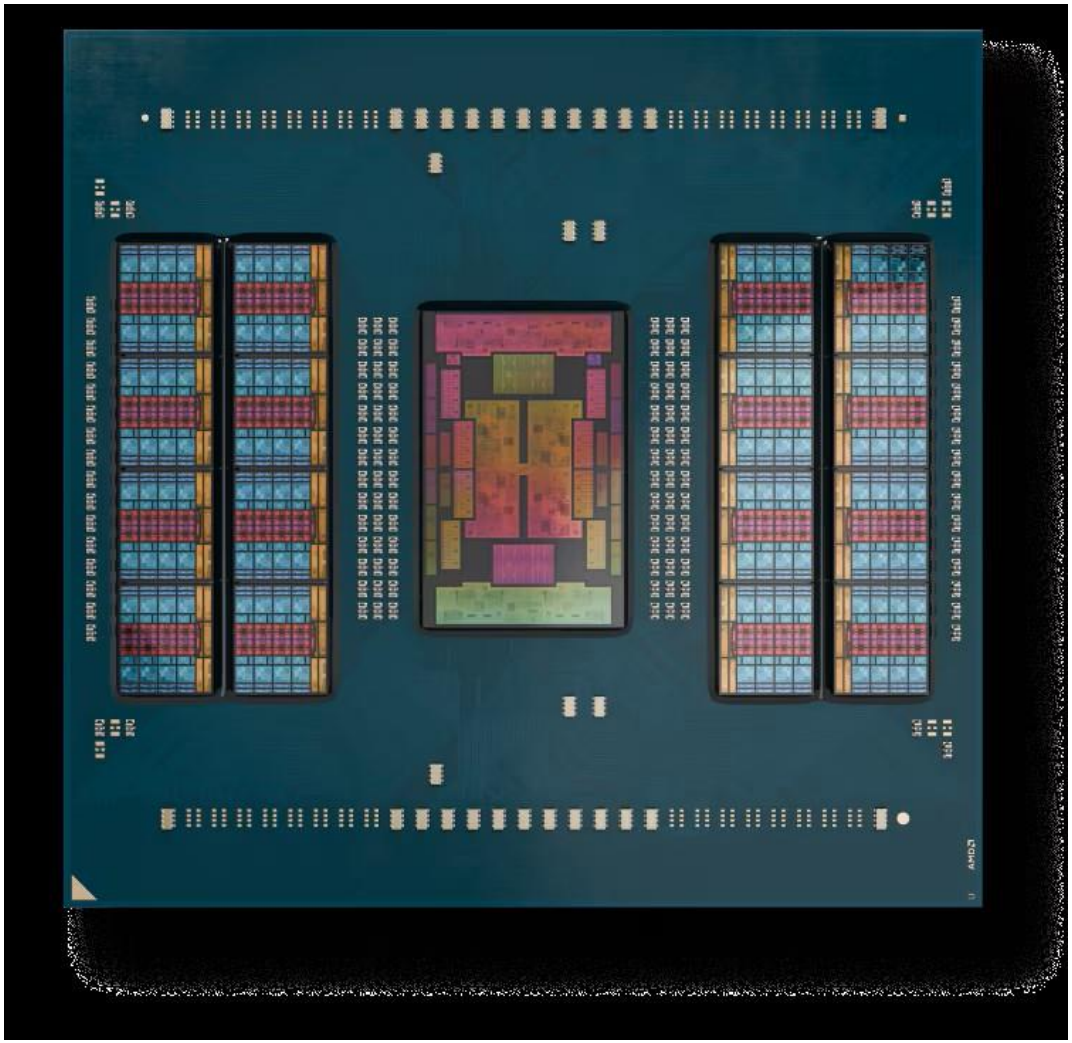
- Large-scale
  - Billion transistor chips
  - Multi-cores, multi-threaded sw
- Power-consumption limited
- Very expensive to design
  - Non recurring engineering costs
  - Migration toward software



Chiplet

A **chiplet** is a small, modular integrated circuit (IC) that serves as a building block in a larger multi-chip module (MCM).

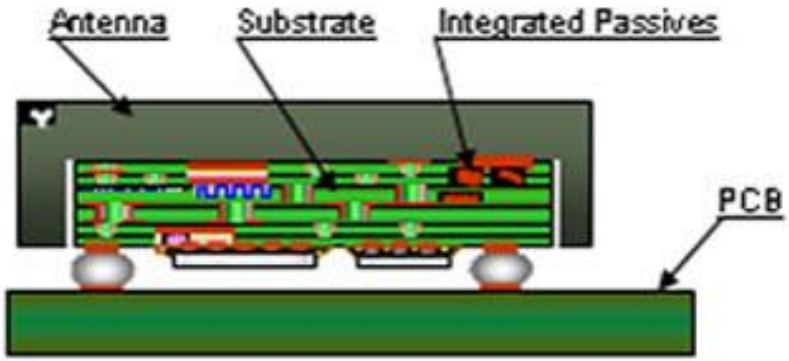
Instead of designing a single, large monolithic chip, manufacturers create multiple smaller chiplets and connect them together using high-speed interconnects.



Zen5
Gen AMD EPYC Processors

Systems in Package (SiP)

- There is a diminishing return in integrating everything on Si
  - Heterogeneous technologies
  - Multiple voltages
  - Thermal issues
- Packaging technology



Schematic diagram of RF-SIP

End, Questions ?

- Design Metrics
- Managing engineering trade-offs

